Statistics Lecture Notes

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Introduction

These are the lecture notes for Math 146 (34644) at Seattle Central College. These notes will be updated throughout the semester. Please note that these are built on top of your assigned readings and it is recommended that you go through those thoroughly first!

Lecture numbers will not exactly correspond to the numbers in the navigation. Here I will keep a running list on what is covered each day.

1. Wed, Sept 30: Went through **Exploring Data** notes.
2. Mon, Oct 5: Went through **Summary Statistics** notes. Also worked on **Topic 1 Check In** (worksheet available with solutions on WAMAP under Additional Class Resources). See additional practice problems at the end of **Summary Statistics** for more practice.
3. Wed, Oct 7: Covered **Examining Relationships**.
4. Mon, Oct 12: Covered **Sample and Study Design**. Additionally completed the **Topic 2 Check In**, which is available on WAMAP.

1 Exploring Data

1.1 Overview

In these notes we will look at a few common methods for visualizing data. There are two main types of data we will be concerned with in this class: **categorical** and **quantitative** data. Categorical data is when the results are descriptive and can be organized into a finite number of answers (or categories). Examples of categorical data include: favorite color, blood type, methods of transportation, favorite food, etc. Quantitative data is data that is numerical *and the numbers are representative of actual amounts*. This can include: weights, heights, times, exam scores, etc.

Example 1.1. Discuss with a partner whether the following are **categorical** or **quantitative**:

- Age
- Birth month
- Birth year
- Hair color
- Zip code
- Number of siblings
- Jersey numbers (in sports)
- Satisfaction score from 1-5

The tools we use to visualize data depend on whether the data is categorical or quantatative.

1.2 Bar and Pie Charts

Bar and **Pie charts** are two ways to visualize the distribution of **categorical** data. Suppose 250 individuals were polled about how they commute to work and the following results were obtained,

Category	Count
Car	134
Public Transit	58

Category	Count
Bike	44
Walk	24

We can visualize this data using either a bar or pie chart.

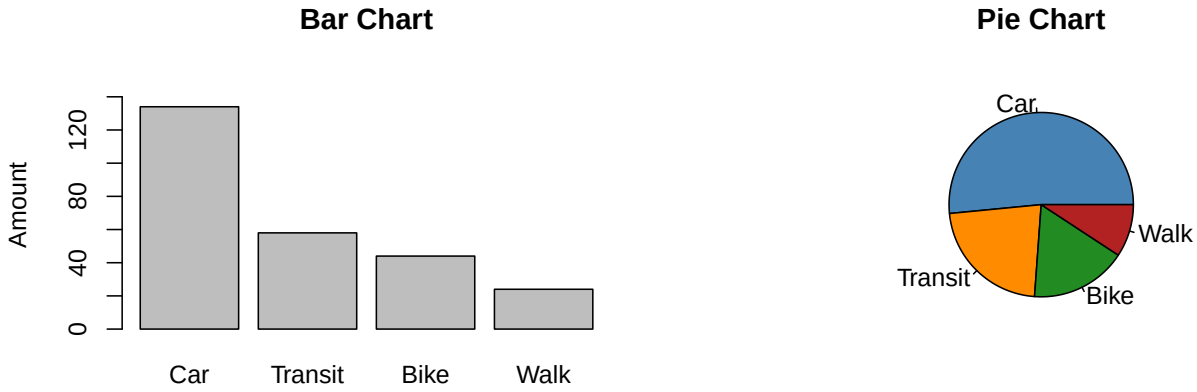


Figure 1.1: The same data shown as a bar chart (left) and a pie chart (right).

Creating a bar chart is simple, each **categorical variable** gets a bar where the height is represented by the number of respondents in the dataset. To create a pie chart, each category gets a slice of the pie proportional to its percent of the responses.

Category	Count	Percentage
Car	134	53.6%
Public Transit	58	23.2%
Bike	44	17.6%
Walk	24	9.6%

Example 1.2. A class of 32 students was surveyed about whether they prefer morning, afternoon, or evening classes. The results are given in the table below.

Time of Day	Count
Morning	10
Afternoon	14
Evening	8

To compute the percentage of votes of each category, we divide the quantity by the total number of students, 32.

Time of Day	Count	Percentage
Morning	10	$10/32 = 31.25\%$
Afternoon	14	$14/32 = 43.75\%$
Evening	8	$8/32 = 25\%$

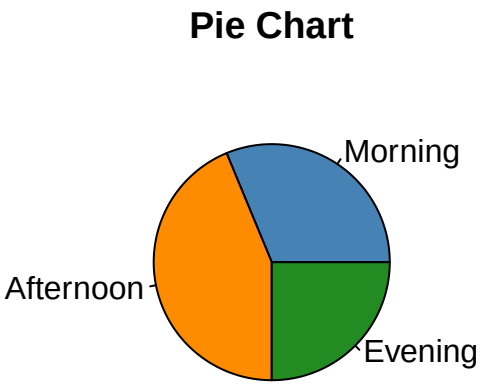


Figure 1.2: A pie chart depicting student class time preferences.

! Picking the Best Visual

In what ways are bar and pie charts similar and in what ways are they different? What are some benefits to a pie chart over a bar chart (and vice versa)?

Answer

A pie chart is better when the main question is about part of a whole, like whether one category is more than half the total.

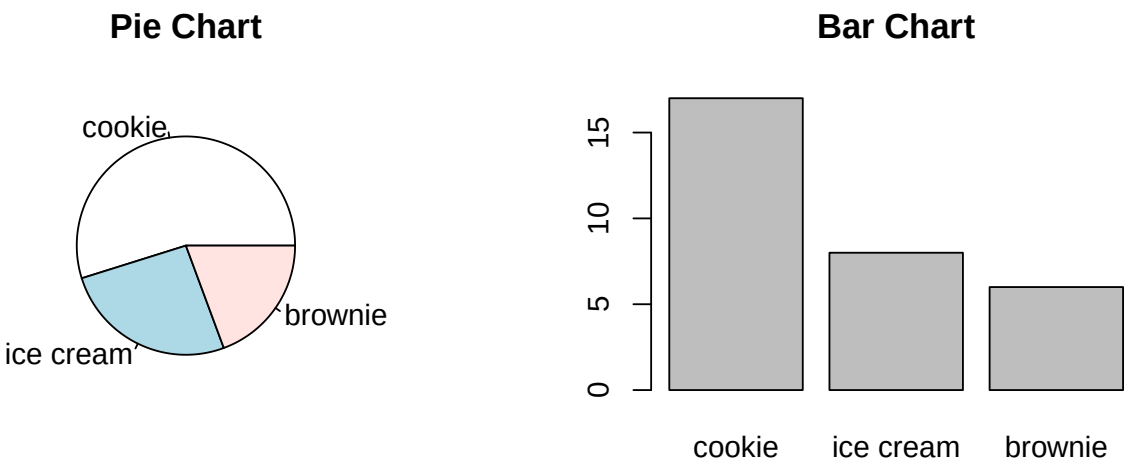


Figure 1.3: Bar and pie charts showing respondents favorite food.

A bar chart is better when we are comparing many close values. For example consider the favorite food of 100 respondents,

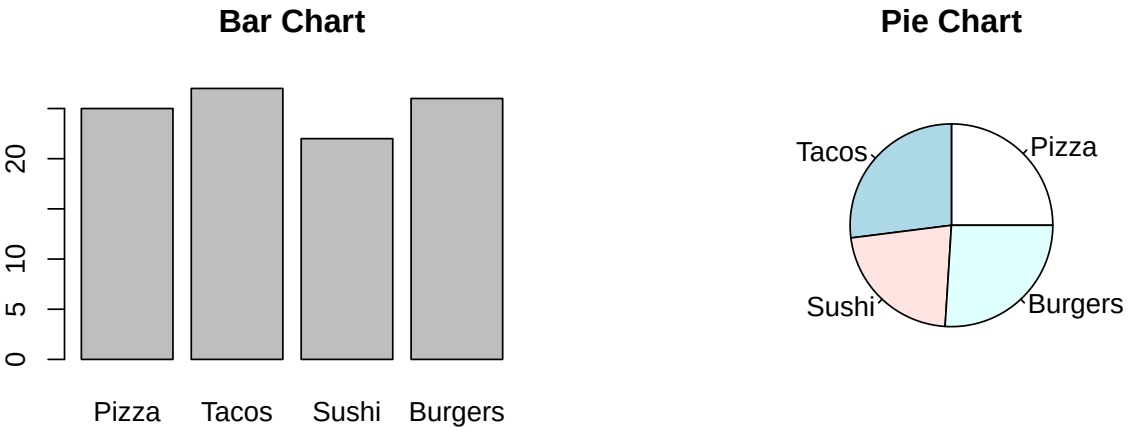


Figure 1.4: Bar and pie charts showing respondents favorite food.

Don't worry so much about creating a pie or bar chart by hand as we will use technology in this class. The below example explain how to create the graphs using StatCrunch.

Example 1.3 (Basic Visuals in StatCrunch)**.** A company has collected data on 500 customers. Open this dataset in StatCrunch. Use StatCrunch to create a bar chart of customer satisfaction by doing the following:

1. Under Graph, select Bar Plot, then With Data.
2. In the pop up screen, select the column SatisfactionRating. Set Type to Frequency and then hit Compute!

Next create a pie chart of the customer regions by selecting Graph > Pie Chart > With Data.

1.3 Misleading Visuals

Some visualizations can be misleading! Sometimes this is just poor judgement, but sometimes this can be done to make a specific point.

Example 1.4 (Bad Visualizations)**.** Can you determine what makes these visuals misleading? Do you think it was done to make a specific point?

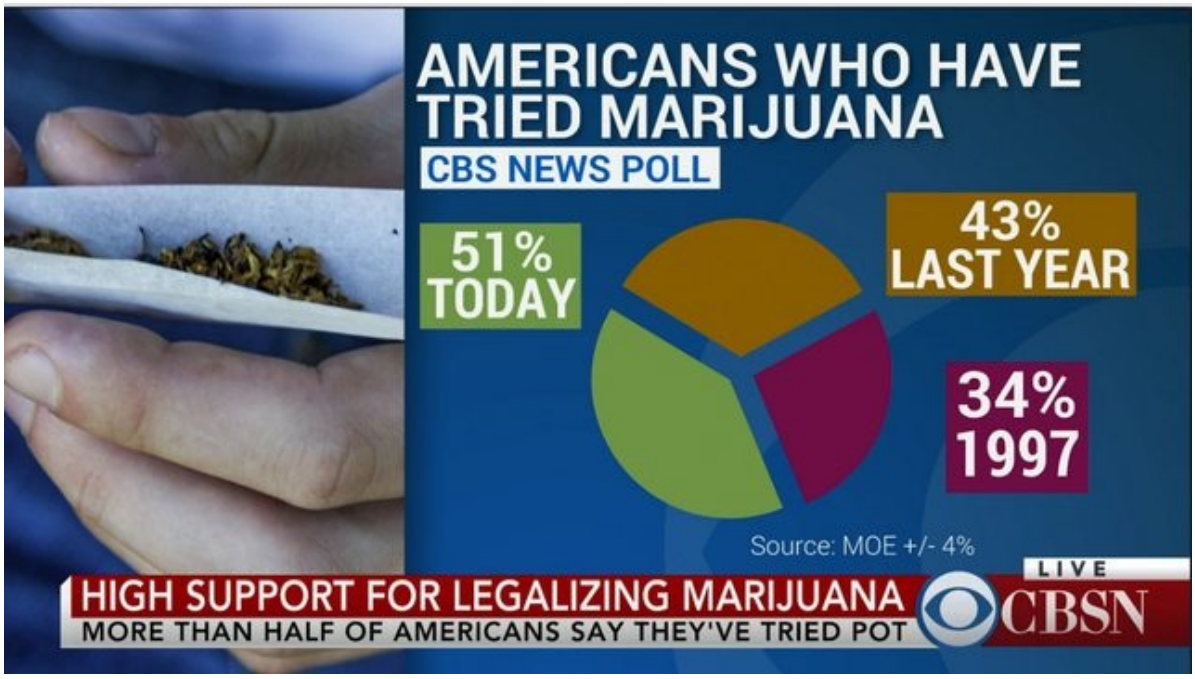


Figure 1.5: Image from www.codeconquest.com/blog/12-bad-data-visualization-examples-explained/

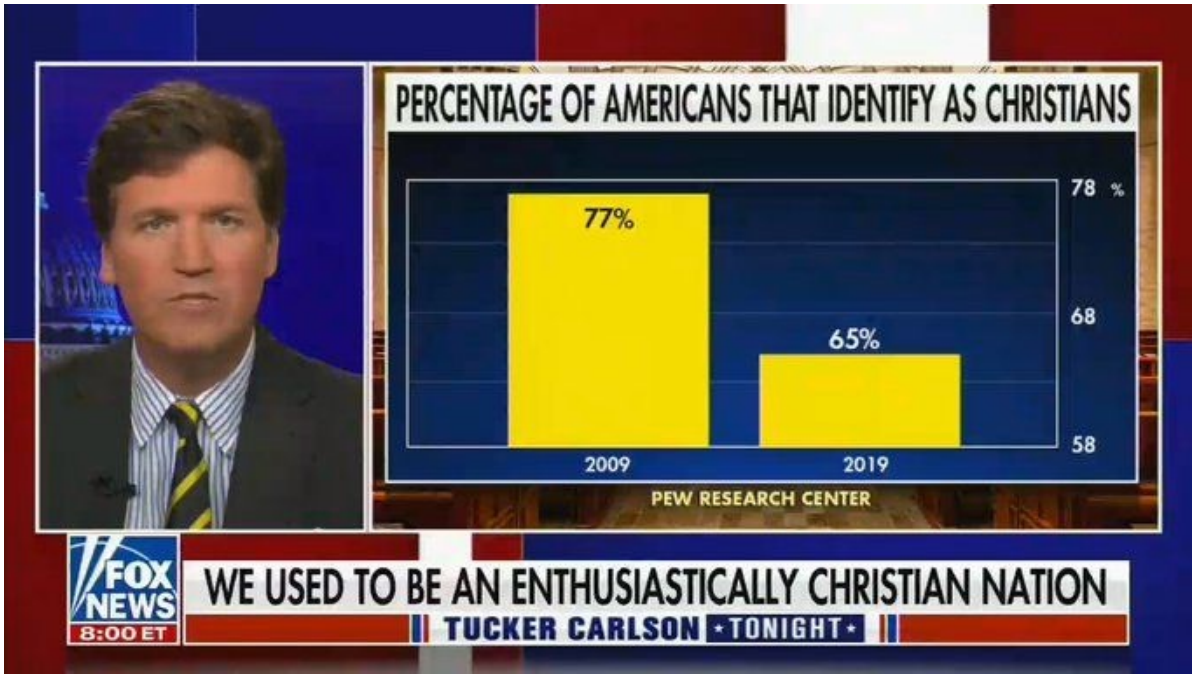


Figure 1.6: Image from www.codeconquest.com/blog/12-bad-data-visualization-examples-explained/

1.4 Histograms

Bar and pie charts are a great way to visualize *categorical* data. Now we want to address ways to visualize **quantitative** data, or data that is in the form of numbers.

Example 1.5 (Quantitative Data)**.** Some example of quantitative data include: test scores, age, height, weight, (systolic) blood pressure, etc.

A good way to look at quantitative data is to use a **histogram**. A histogram breaks the range of all possible values into smaller intervals, then counts how many observation fall into each window. This looks kind of like a bar chart, but the bars sit right next to each other.

Example 1.6. Students in a calculus class obtain the following scores on the midterm exam,

$$100, 85, 91, 76, 88, 82, 95, 62, 84, 79, 90, 73, 87, 81, 92, 77, 86, 83, 80, 89, 74, 93, 85, 79, 82$$

The lowest score is a 62 and the highest score is 100. Let's break down the range $[60, 100]$ into intervals of length 5.

Interval	Count
(60,65]	1
(65,70]	0
(70,75]	2
(75,80]	5
(80,85]	7
(85,90]	5
(90,95]	4
(95,100]	1

From this we can make a histogram.

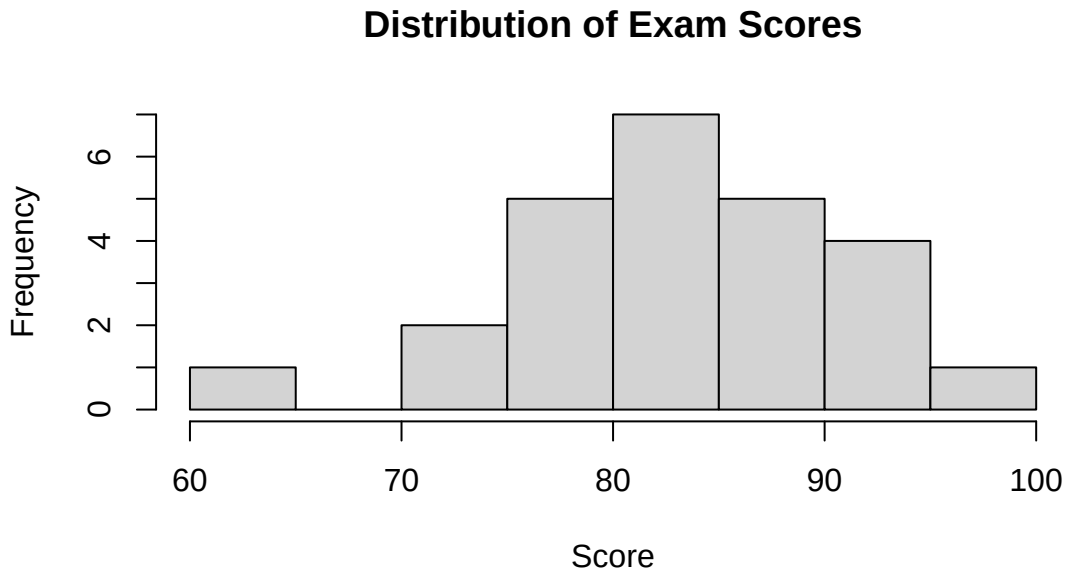


Figure 1.7: A histogram of test scores.

It is typical to call the intervals of a histogram **bins**, although the reading on WAMAP will call them **classes**. A histogram should always have bins of the same length. It can be misleading to use different bin lengths in the same way it is misleading to use breaks on the vertical axis of bar charts.

! Histogram Versus Bar Chart

Histogram "bars" touch, while bar chart bars do not. Why do you think this distinction is made?

Answer

Histogram bars touch because the bins cover a continuous range of numbers with no gaps between them. Bar chart bars are separated because each bar is a distinct category, and there's nothing connecting the categories together.

Most computer programs will set automatic bin sizes based on the distribution of the data you get it, although you will typically have an option to manually input bin specifications. In StatCrunch you are give the optional inputs of the starting value and the bin width.

Bin Endpoints

For whatever computer program you use, make sure you know how the **endpoints** of your bin intervals are treated. In Example Example 1.6, I wrote intervals like (70,75] which means 70 is excluded from the bin and 75 is included in the bin. The statistical programming language R (what generates the graphs in these notes) uses right inclusive bins, however StatCrunch and Google Sheets use *left* inclusive bins.

The histograms below represent *the same* data, just with different bin sizes.

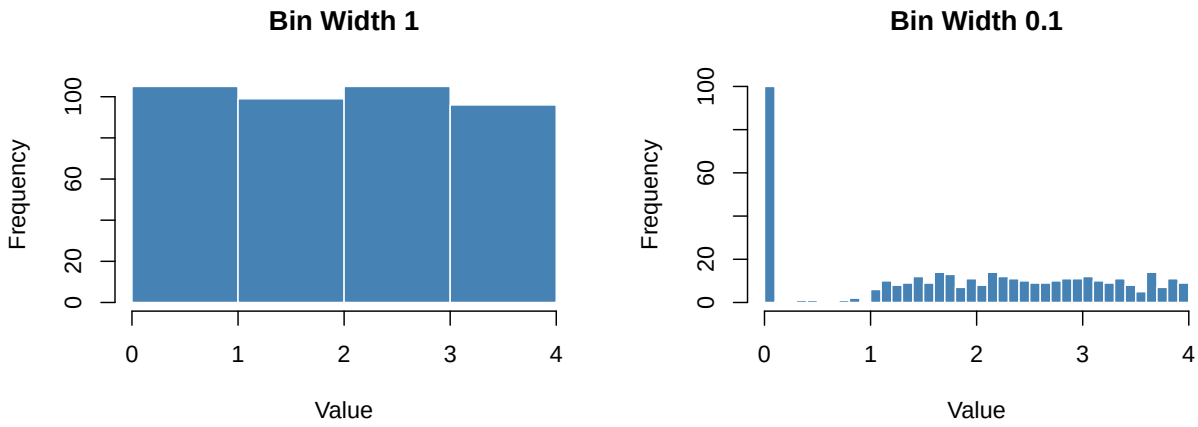


Figure 1.8: Histograms of the same dataset with different bin widths.

Example 1.7. Discuss the following with a partner or two:

- Which histogram do you think is better for the above data set and why?
- What goes wrong if bins are too wide? What if bins are too narrow?
- Is there one "correct" bin width?
- Could someone manipulate bin size to mislead the public about a dataset?

Example 1.8 (Histograms on StatCrunch)**.** Reopen the customer dataset in StatCrunch. Create a histogram of customer ages by going to Graph > Histogram.

Example 1.9 (High Tide Heights)**.** The following histogram shows 60 high tide heights over the month of July at a beach. What **percentage** of high tides were more than 2 meters?

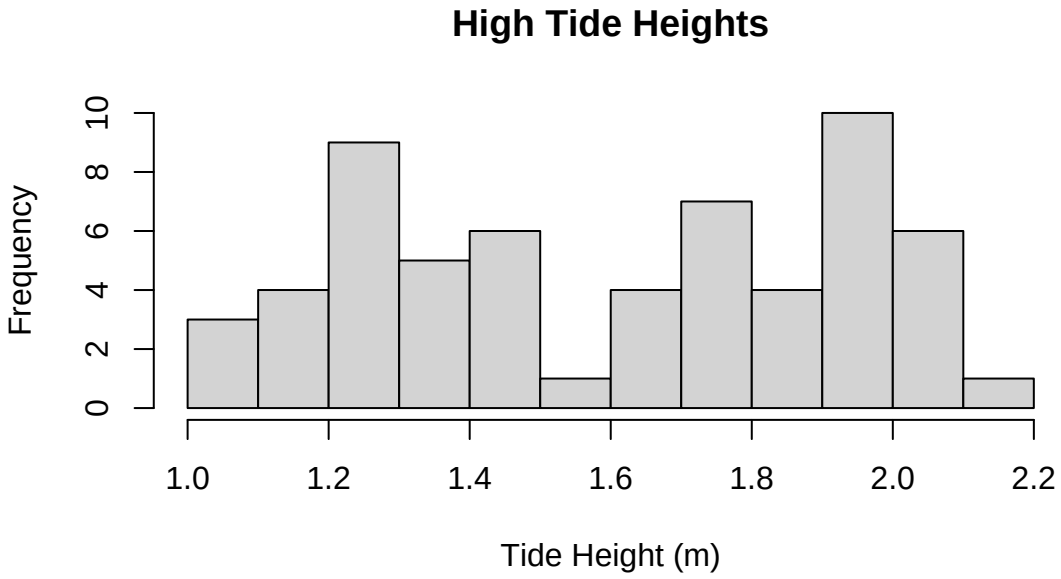


Figure 1.9: Histograms of high tide heights in July.

Answer

We want to compute,

$$\frac{\text{high tides over 2 m}}{\text{total number of high tides}} \cdot 100,$$

where multiply by 100 turns the decimal into a percent. The total number of high tides is 60 as stated in the problem. If not given explicitly, we could carefully sum the quantity in each bin.
The number of high tides that were more than 2 meters is given by the total in the last two bins, $6 + 1 = 7$.

$$\frac{7}{60} \cdot 100 \approx 11.67\%$$

Information Lost

What kinds of information do we lose by only seeing a histogram of the data?

We can use words to descibe the shape of distributions. Some common terms are:

- A distribution is **symmetric** if the left and right sides are roughly mirror images of each

other.

- A distribution is **skewed right** if it has a long tail stretching to the right. Most values are low, with a few unusually high values.
- A distribution is **skewed left** if it has a long tail stretching to the left. Most values are high, with a few unusually low values.

The direction of the skew is the direction of the *tail*, not the direction of the peak. A skewed-right distribution has its peak on the left.

We also describe distributions by their number of peaks. A distribution with one peak is **unimodal**, and a distribution with two clear peaks is **bimodal**. The symmetric and skewed examples below are all unimodal.

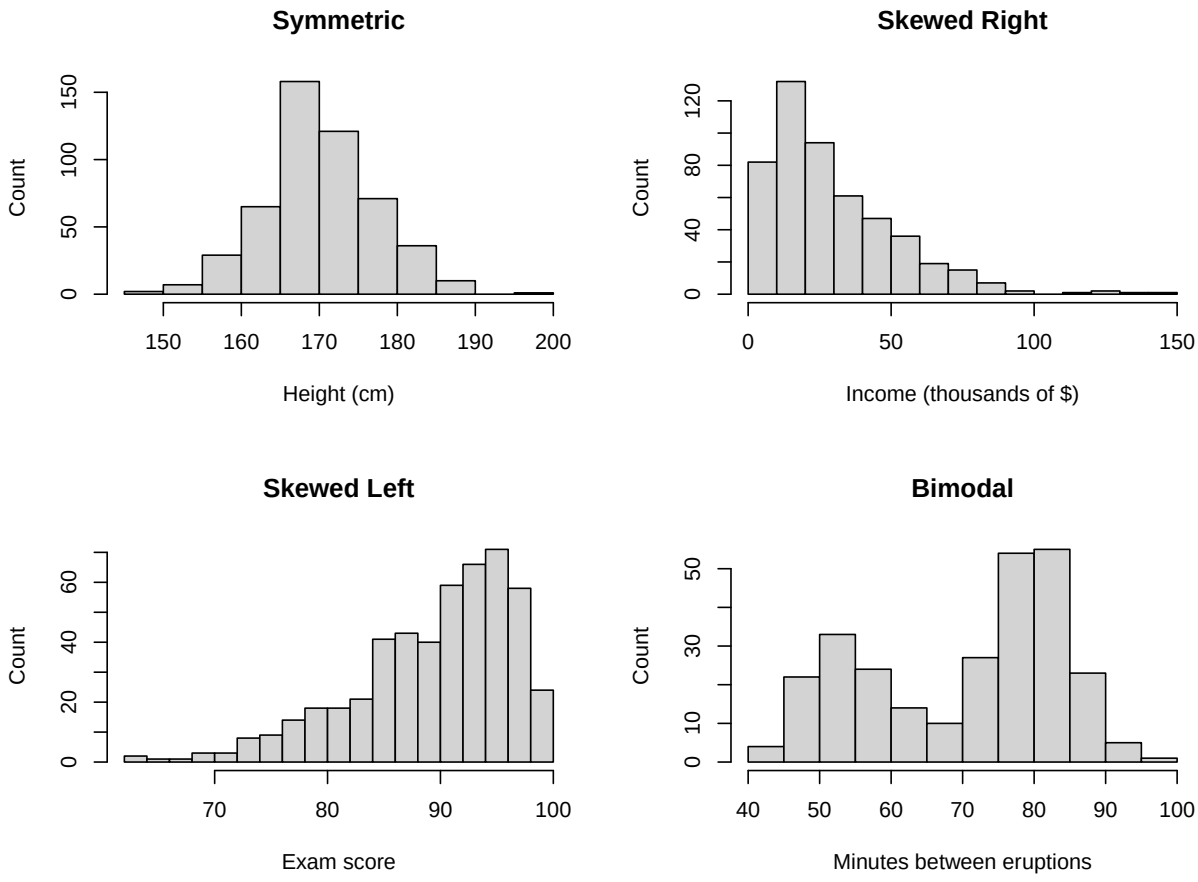


Figure 1.10: Four common distribution shapes.

Shape	Real World Examples
Symmetric	Adult heights (for one sex), repeated measurements of the same quantity, standardized test scores
Skewed right	Household income, home prices, number of social media followers, time spent waiting in line
Skewed left	Scores on an easy exam, age at death in countries with high life expectancy
Bimodal	Customers in a restaurant, shoe sizes of a population not separated by gender

! Why Bimodal?

A bimodal distribution often means the data contains two different groups mixed together. For the restaurant example, what are the two groups? Can you think of another variable that might be bimodal?

Example 1.10 (Plant Heights)**.** A biology class grew 40 bean plants under identical conditions and measured each plant's height, in centimeters, after six weeks. The histogram below shows the results.

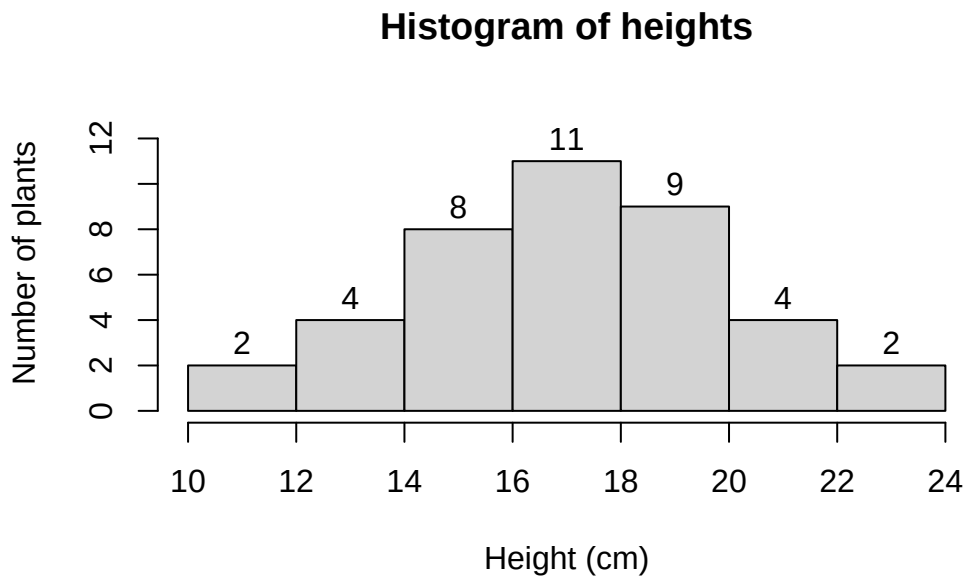


Figure 1.11: Histogram of bean plant heights after six weeks.

1. What is the bin width of this histogram?

2. Describe the shape of the distribution. Around what height is the center?
3. What percentage of plants grew taller than 20 cm?
4. What percentage of plants had heights between 14 and 20 cm?
5. Can you tell from the histogram how tall the shortest plant is?

Answer

1. Each bin covers 2 cm, for example 16 to 18 cm.

2. The distribution is roughly **symmetric**: the tallest bar is in the middle, at 16–18 cm, and the bars decrease at about the same rate on both sides. The center is around 17 cm.

3. The last two bins contain plants taller than 20 cm: $4 + 2 = 6$ plants.

$$\frac{6}{40} \cdot 100 = 15\%$$

4. The bins from 14 to 20 cm contain $8 + 11 + 9 = 28$ plants.

$$\frac{28}{40} \cdot 100 = 70\%$$

5. No. We only know the shortest plant is somewhere between 10 and 12 cm. The histogram does not show individual values.

2 Summary Statistics

We've given a few examples of ways to visualize data, now we want to talk about some important **summary statistics**. Summary statistics are quantitative values one can use help describe the shape of a distribution. We will talk about measures of **center** and measures of **spread**.

Almost always, summary statistics can only be used on *quantitative data*.

2.1 Measures of Center

The goal of measures of *center* is to describe the value of a typical datapoint. There are three measures of center we will cover:

- **Mode**: The value(s) with the highest frequency in the dataset. Mode is the *only* measure of center that can be used with categorical data.
- **Mean**: The average of the values. If a dataset has n values x_1, x_2, ..., x_n, then the mean $\bar{x}$ is given by

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

- **Median**: The midpoint of the distribution. It is the number such that half the observations fall above and half fall below.

Example 2.1 (Classroom Ages)**.** A college class has 22 students with the following ages: 18, 22, 21, 21, 21, 18, 20, 42, 23, 44, 18, 18, 21, 18, 22, 18, 22, 23, 20, 38, 20, 34. Compute the mean, median, and mode.

See this data on StatCrunch to perform the computations there.

Answer

To compute the mode and median it helps to order the data from smallest to largest.

$$18, 18, 18, 18, 18, 18, 20, 20, 20, 21, 21, 21, 21, 22, 22, 22, 23, 23, 34, 38, 42, 44$$

The mode is 18, which occurs in the dataset 6 times.

Since the dataset is 22 students, the median is the value in-between the 11th and 12th observations. The 11th and 12th observation both happen to be 21, therefore the median is 21.
The mean is,

$$\frac{6 \cdot 18 + 3 \cdot 20 + 4 \cdot 21 + 3 \cdot 22 + 2 \cdot 23 + 34 + 42 + 44}{22} = \frac{522}{22} = 23.73.$$

Notice that the mean is slightly larger than the median this is because there are a handful of older students with *outlier* ages. The mean gets pulled by outliers more than the median does.

! Median on Datasets of Even Size

For this course, to compute the median when the dataset has an *even* number of observations average the two observation in the middle. This is a typical convention chosen in statistics.

The difference between the median and mean of a distribution can reveal information about the shape of the distribution. In particular, if a distribution is **skewed** or **symmetric**.

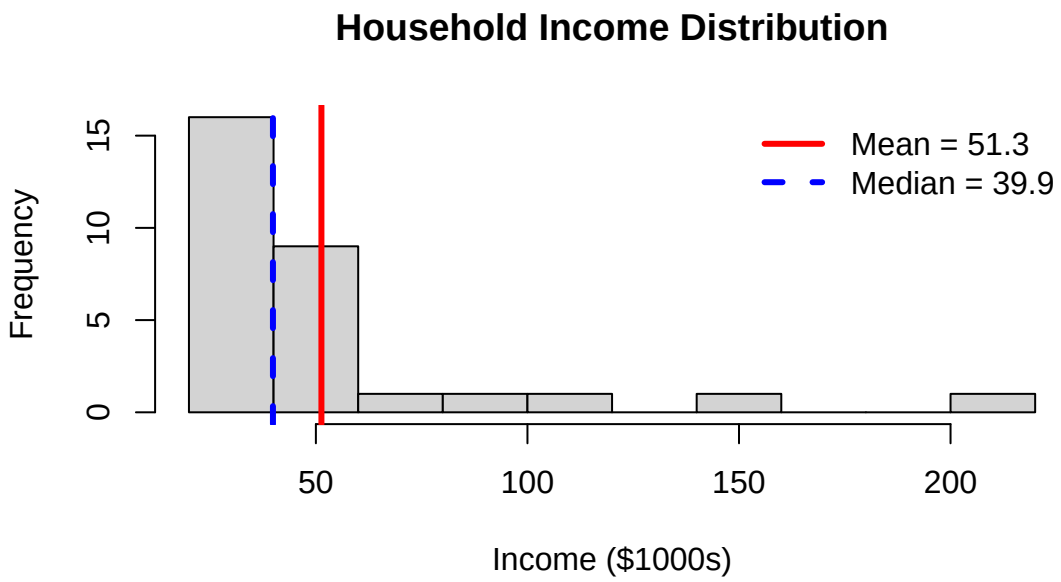


Figure 2.1: Right-skewed household income distribution.

When a distribution is skewed, the mean is pulled more towards the **tail** than the median.

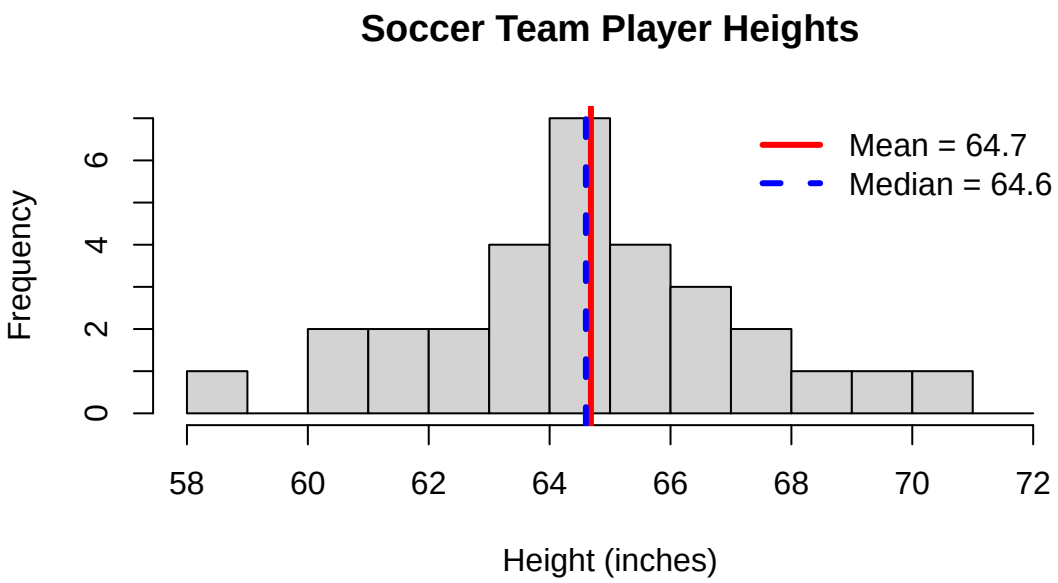


Figure 2.2: Approximately symmetric distribution of soccer team heights.

! Mean Versus Median

The reason **outliers pull the mean more than the median** is because mean minimizes the error *squared*, while the median minimizes the absolute error. Squaring the error means outliers will have extra impact. This fact is outside of the scope of this course, but I am happy to chat about it for curious minds!

Example 2.2 (Creating Data)**.** Using StatCrunch complete the following:

1. Create a dataset of 10 student ages where the median is 20 and at least one student is not 20.
2. Is it possible to do this and have no student be 20 years old?
3. If we increased the size of the dataset to 15 students can you still create a dataset with median 20 *and* no students are 20 years old?

Answer

1. One possible answer: 18, 18, 18, 19, 20, 20, 20, 21, 22, 23.
2. Yes it is! Since the dataset has an *even* size, we can make it so the 5th entry is 19 and the 6th entry is 21. Such as: 18, 18, 18, 19, 19, 21, 22, 23, 25.
3. This is **not** possible if the dataset has an *odd* size. When there are 15 students, the

age of the 8th youngest student is the median–so that student must be 20.

Example 2.3.

1. Using StatCrunch or Google Sheets, create a dataset of 25 non-negative integers (e.g. whole numbers 0 and larger) where the mean is less than the median. Plot the histogram of the data.
2. (Challenge) If I have a dataset of 25 **non-negative integers** and the median is 50, what is the smallest possible mean?

Answer

1. Your dataset should be **skewed to the left** in this example.
2. The smallest possible mean occurs when 12 data points have value 0 and 13 have the value 50. This guarantees the median is 50. So the mean is

$$\frac{12 \cdot 0 + 13 \cdot 50}{25} = 26.$$

We can't make any of numbers smaller or else we would either change the median or break the non-negative rule. Making any of the numbers bigger would increase the mean!

When to use mode?

Discuss: what might be a situation where the mode is the most relevant summary statistic to use?

Tip

Mode is the only measure of center that can be used for categorical data. Also, mode is great for bimodal (or multimodal) distributions!

Nobody's Average

In 1950 the airforce measured 140 physical dimension of 4063 pilots and designed the cockpit to match he average (mean) dimensions. Lieutenant Gilbert S. Daniels then compared this "average pilot" to individual pilot records to show that no pilot was within 15% of the average on the ten most relevant dimensions. In other words, the cockpits were designed to fit no one! Thus lead to the adaptation of adjustable seats.

2.2 Measures of Spread

Example 2.4 (Commute Times)**.** Data on StatCrunch. The below histograms depict the commute time of 500 people who work in Town A and Town B. How are the datasets similar and how are they different?

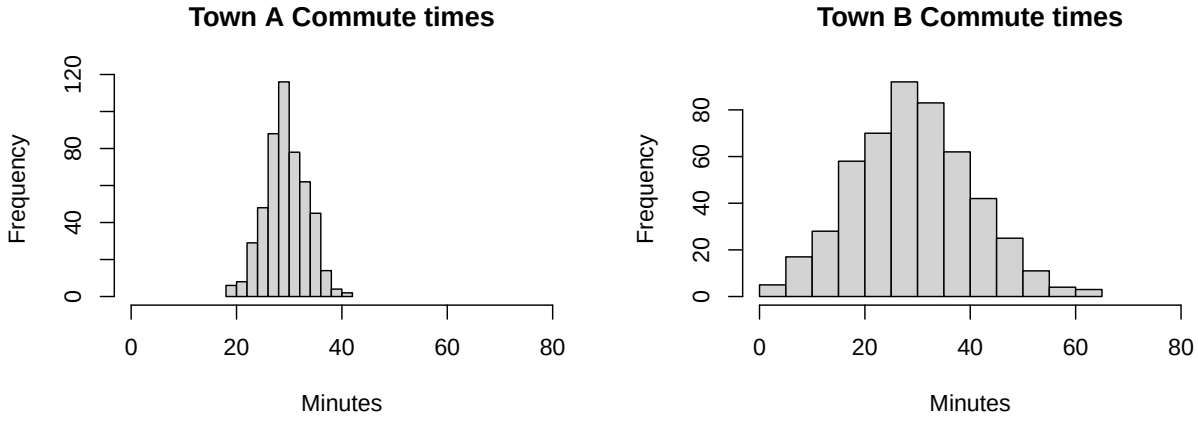


Figure 2.3: Commute times in two different towns.

The above example illustrates that it is important to have statistics that describe the *spread* of a distribution. There are a few we will investigate today.

2.2.1 Range

The **range** of a dataset is the largest observation minus the smallest observation.

Example 2.5. The commute times data is given here. Compute the range of the commuting data for each town.

Answer

For Town A, the longest and shortest commute times are 42 and 18 minutes, respectively. So the range for Town A is $42 - 18 = 24$.
For Town B, the longest and shortest commute times are 62 and 1 minute, respectively. So the range for Town B is $62 - 1 = 61$.

Range does give us an initial look at the spread of the distribution, but **outliers** can obfuscate how range relates to spread. Consider two classes with the following histograms of exam scores:

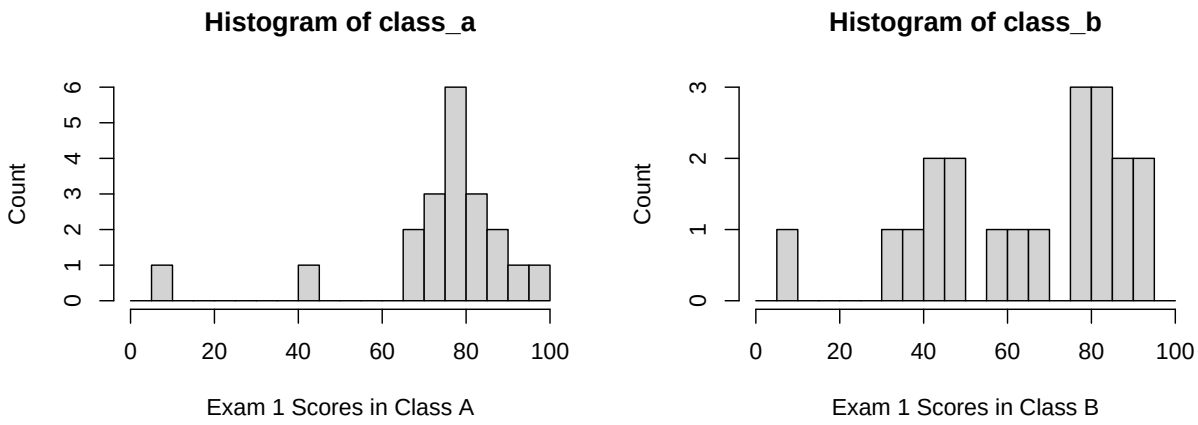


Figure 2.4: Scores on Exam 1 in Class A and Class B

The range of both distributions is (approximately) the same, but the spread looks very different. Only 2 out of 20 students scored below a 60 in Class A, while 8 students (out of 20) scored below a 60 in Class B.

2.2.2 Inter-Quartile Range (IQR)

Recall that the **median** divides the observations into two parts of equal size. This is because we defined the median as the value which half the observations are below and half are above. **Quartiles** divide the data into 4 equal parts. Specifically we define:

- The first quartile (or lower quartile), Q1, is the value such that 25% of the data lies below this value.
- The second quartile (or median), M, is value where 50% of the data lies below this value.
- The third quartile (or upper quartile), Q3, is the value such that 75% of the data lies below this value.

Computing the min, Q1, Q2, Q3, and max of a data set is known as the **five-number summary**.

> **!** Important
>
> Note that Q1 and Q3 are computed differently on Google Sheets than on StatCrunch (and in this class). While they are effectively the same, it will sometimes cause incorrect homework answers.

Example 2.6. Below are the exact list of Exam 1 scores for Class A and Class B (from the histograms in Figure Figure 2.4). Compute the five-number summary for each class. What differences do you observe about the distribution from the statistics?

Class A	Class B	Position Number
10	10	1
44	34	2
66	36	3
70	44	4
71	45	5
73	48	6
74	50	7
76	56	8
77	64	9
79	66	10
80	76	11
80	77	12
80	80	13
82	82	14
83	83	15
83	84	16
86	88	17
87	89	18
91	91	19
96	95	20

Answer

Each class list is already ordered for us, so we have:

- Min = 1st grade –> Class A = 10, Class B = 10
- Q1 = between 5th and 6th grade –> Class A = 72, Class B = 46.5
- M = between 10th and 11th grade –> Class A = 79.5, Class B = 71
- Q3 = between 15th and 16th grade –> Class A = 83, Class B = 83.5
- Max = 20th grade –> Class A = 96, Class B = 95

The **inter-quartile range** (IQR) is defined as Q3-Q1.

Example 2.7. Compute the IQR for the exam scores by class.

Answer

For Class A, IQR = Q3 - Q1 = 83 - 72 = 11. For Class B, IQR = 83.5 - 46.5 = 47.

! IQR Versus Range

Why might IQR be seen as a better gauge of spread than range?

2.2.3 Outliers

An **outlier** is an oberservation that lies far away from the general distribution. The readings define outliers as points that are:

- Smaller than $Q1 - 1.5 \cdot (IQR)$, or
- Larger than $Q3 + 1.5 \cdot (IQR)$.

Do not think of this as a set in stone rule. In fact, the multiplier 1.5 was not chosen with any real mathematical theory behind it. Use these values to flag *potential* outliers in the dataset, but context is an important factor when determining what is an outlier. Moreover, **do not** automatically remove outliers from a dataset

Example 2.8 (Bimodal Commute Times)**.** We have the following list of commute times for a neighborhood:

24	25	26	27	27	28
28	29	29	30	30	30
31	31	32	32	33	33
34	34	35	35	36	36
37	38	39	40	41	42
72	74	76	78	80	82

Let's start by looking at the five-number summary.

```
[1] 24.0 29.5 33.5 39.5 82.0
```

From this we can compute the lower and upper fence:

$$\text{lower} = 29.5 - 1.5(10) = 14.5, \quad \text{upper} = 39.5 + 15 = 54.5$$

That means 6 data points are flagged as outliers using this rule: 72, 74, 76, 78, 80, 82. Let's look at the histogram of the distribution,

What we see from the histogram is a **bimodal** distribution. There is one subset of the neighborhood that commutes a shorter distance (think across town), while a small subset of the neighborhood commutes to a different city. This is an example where the $1.5 \times$ IQR rule is a bad fit.

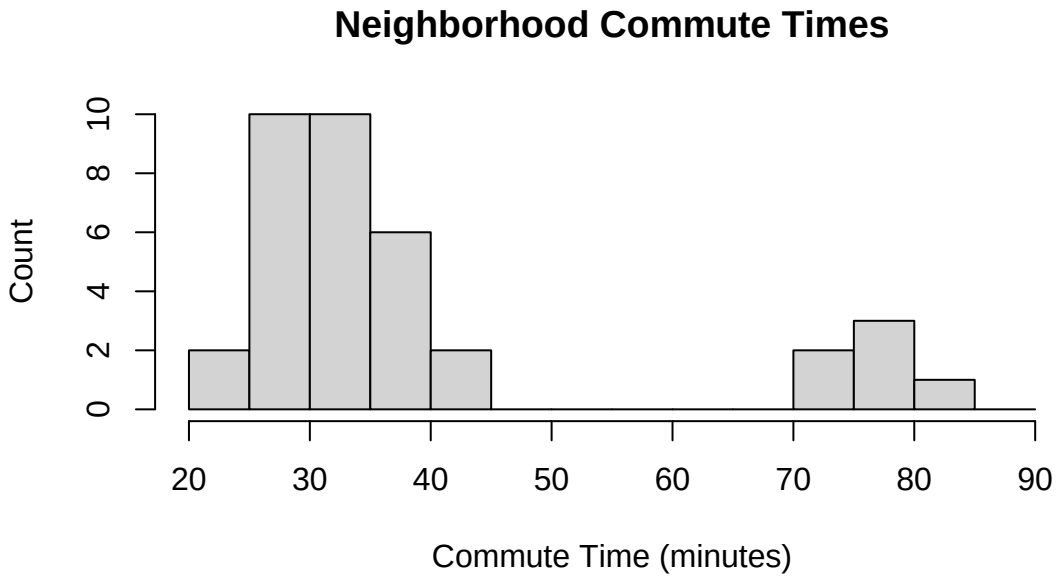


Figure 2.5: Histogram of neighborhood commute times.

! Removing Data

There is a false notion that outliers should be removed from the data set but this is **not true**! Outliers should be expected in data and are part of the underlying distribution. Only remove data points if they are the result of mistake in data entry or collection, or if they can be explained by a fundamentally different process which you are not interested in investigating.

2.3 Boxplots

Boxplots are a way to visualize the five number summary as well as outliers. Let's see this with an example.

Example 2.9 (Boxplots in StatCrunch)**.** For this example use this data on StatCrunch. To make a boxplot of the data select Graph > Boxplot. Just select annual_income and then compute.

The box of the boxplot goes from Q1 to Q3 and the median is shown as a horizontal line. The two sticks (called whiskers) on either side of the box go until the largest data point *that isn't an outlier* and the stars (called fences) indicate outlier values. If we deselect the box indicating "Use fences to identify outliers" in StatCrunch then the whiskers will go all the way until the end of the range.

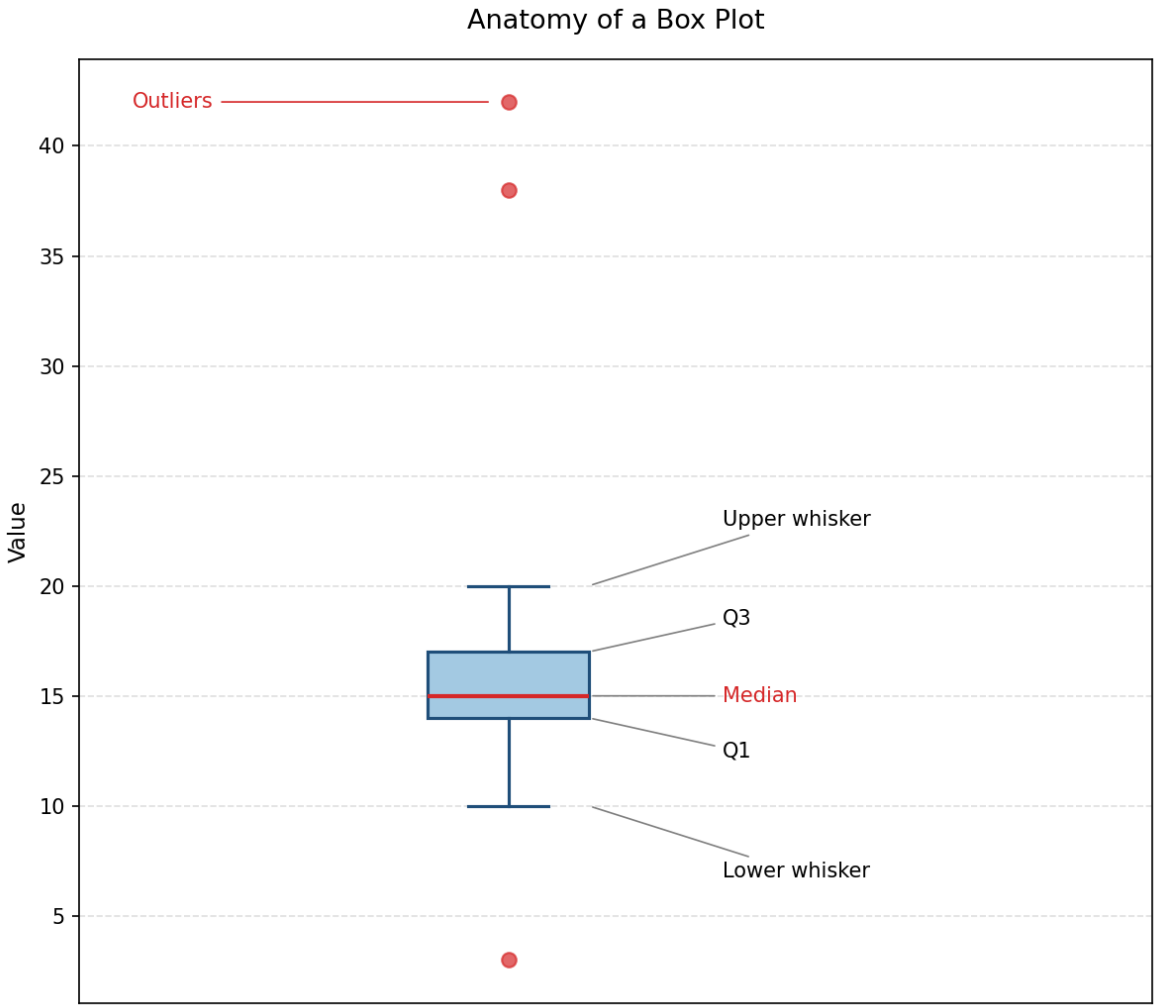


Figure 2.6: Labeled boxplot.

Example 2.10. The boxplots below show the typed words per minutes for support agents and software engineers. Compare the two distributions using what you know about boxplots.

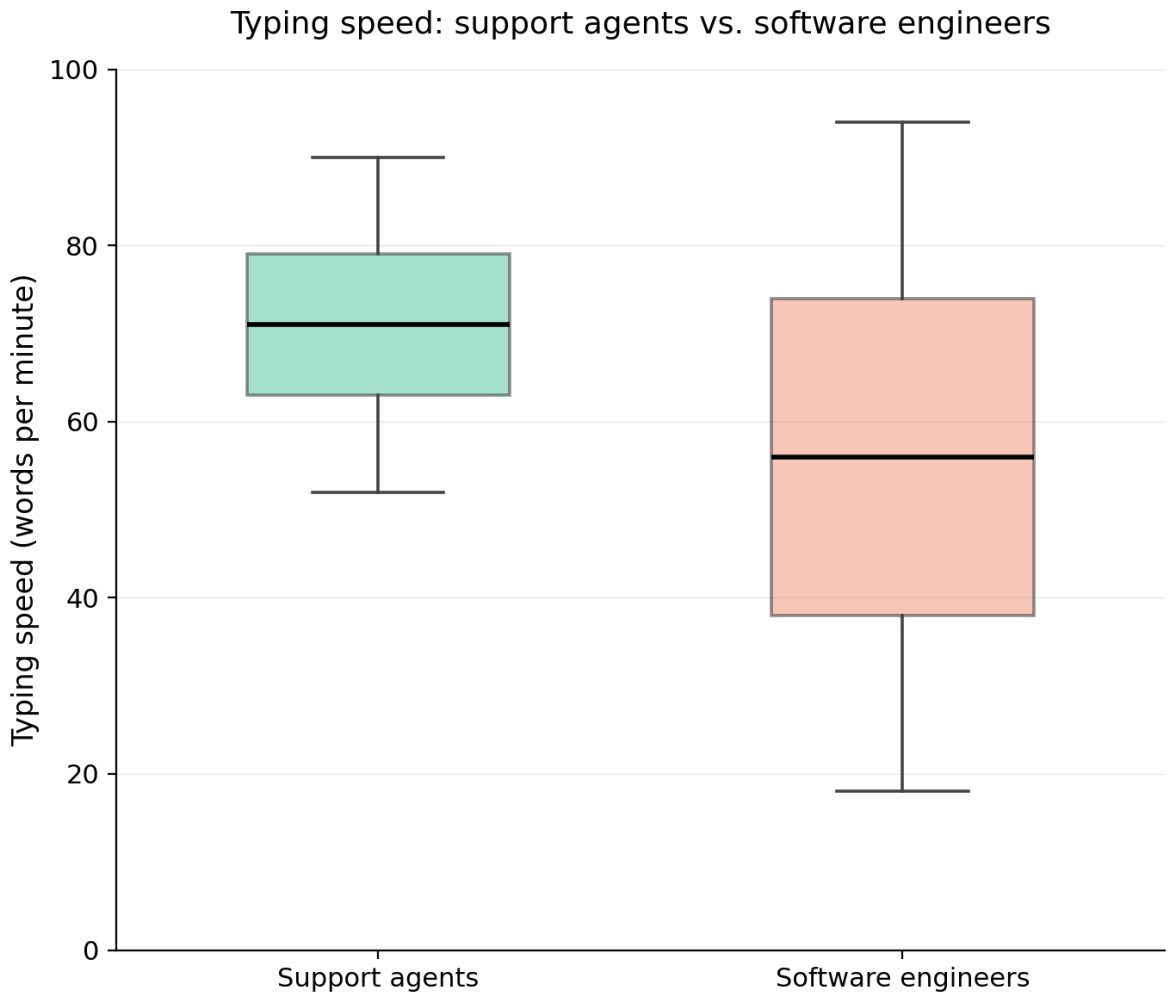


Figure 2.7: Typing speed boxplots for support agents and software engineers.

Answer

The median support agent is a faster typer than the median software engineer. Software engineers have a much larger *variation* in typing speeds then support agents.

2.4 Standard Deviation

Here we will define and use the **sample standard deviation**. This is slightly different, but related to, the *population* standard deviation as well as the variance. These are terms we will

define explicitly later. Given data points x_1, ..., x_n the **sample standard deviation** is defined by the formula

$$SD = \sqrt{\frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n-1}}$$

where $\bar{x}$ denotes the mean of the dataset. Let's see this formula in action with a small example,

Example 2.11. Anna denotes how many Reese's Peanutbutter Cups her 10 friends eat at a party. She gets the following amounts:

$$3, 4, 3, 6, 6, 2, 1, 8, 5, 5.$$

Compute the mean and standard deviation of the dataset.

Answer

Let's compute the mean,

$$\bar{x} = \frac{3+4+3+6+6+2+1+8+5+5}{10} = \frac{43}{10} = 4.3.$$

Next compute the differences,

$$-1.3, -0.3, -1.3, 1.7, 1.7, -2.3, -3.3, 3.7, 0.7, 0.7.$$

Square and sum,

$$1.69 + 0.09 + 1.69 + 2.89 + 2.89 + 5.29 + 10.89 + 13.69 + 0.49 + 0.49 = 40.1$$

Now divide by $n - 1 = 10 - 1 = 9$ and take the square root,

$$SD = \sqrt{\frac{40.1}{9}} = \sqrt{4.456} \approx 2.11.$$

We claim that standard deviation is telling us *something* about the spread of the distribution, but we can be more precise than that. The standard deviation tell us the average distance from an given datapoint to the mean. A larger standard deviation means the data is more spread out, while a smaller standard deviation tells us the data is compact.

Example 2.12 (Standard Deviation in StatCrunch)**.** For this example use this data on StatCrunch. Compute the standard deviation of the income dataset.

Example 2.13. Which histogram has the smallest standard deviation? The largest?

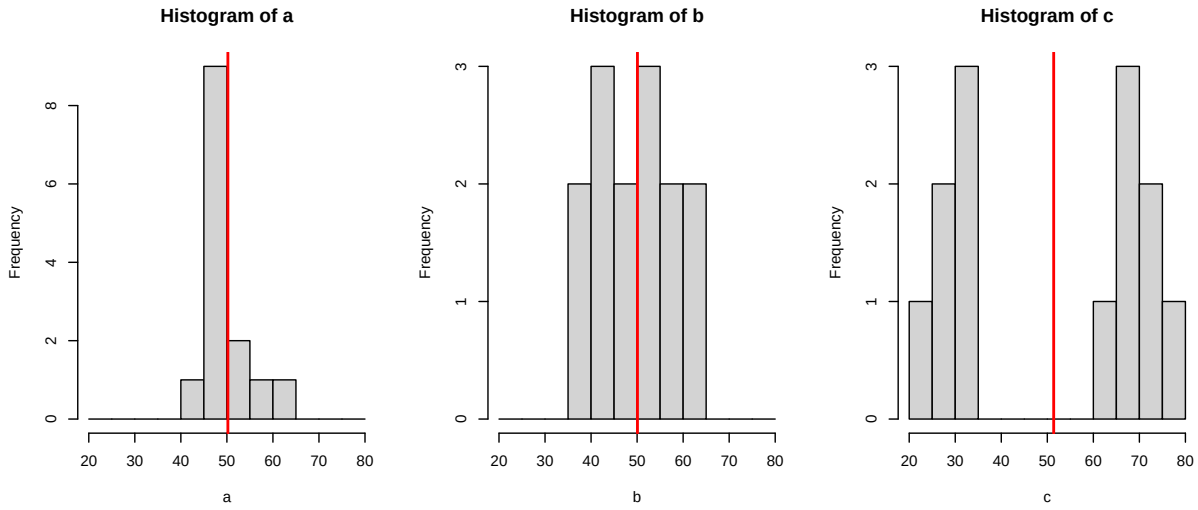


Figure 2.8: Histograms of data with similar mean and different standard deviations.

2.5 Normal Distribution and the 68-95-99.7

Quantitative data is **normally** distributed when its distribution looks like a symmetric "mound" or "bell" shape.

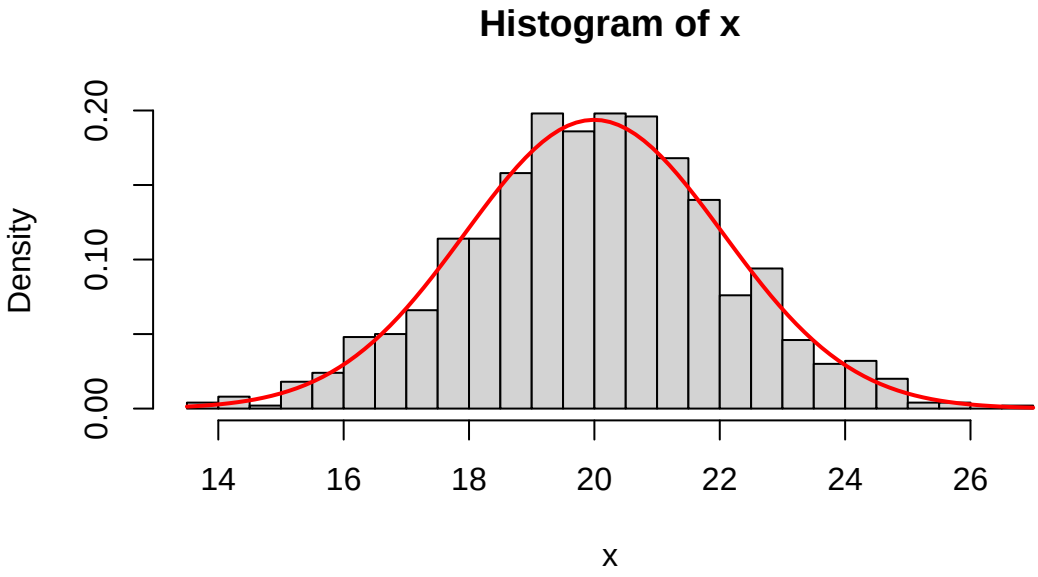


Figure 2.9: Histogram of approximately normal data.

Things that are approximately normal in distribution include:

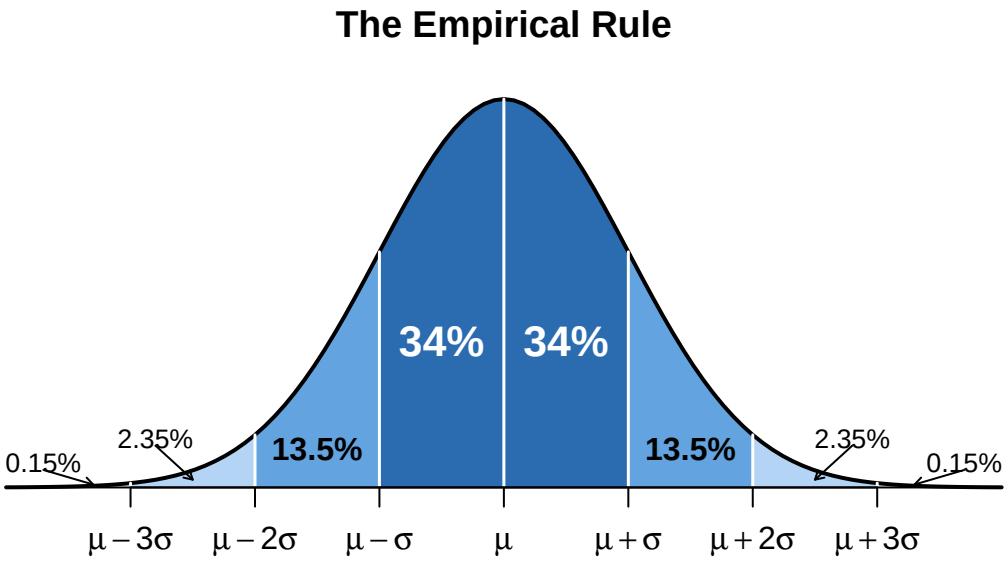
- Baby birth weights
- Heights of adults by sex
- Blood pressure of adults
- Standardized testing scores

Statistics over time may have you thinking everything is normal, but this is not true! You should always question whether a distribution is normal or has some other distribution. Things that are *not* normal include:

- Income
- Time between earthquakes in a given region
- Number of shots taken in a soccer game
- Amount of time someone waits in a line

If a distribution is normal though, the **empirical rule** (or 68-95-99.7 rule) gives us a rough estimate of what percentage of the data falls withing 1, 2, and 3 standard deviations of the mean. We have:

- 68% of the data falls within 1 standard deviation of the mean
- 95% of the data falls within 2 standard deviations of the mean
- 99.7% falls within 3 standard deviations of the mean



Example 2.14 (Baby Weights)**.** The mean birth weight at a hospital is 7.3 lbs with a standard deviation of 1.1 lbs. Assume that the birth weight is approximately normally distributed.

1. What percentage of babies weigh between 6.2 lbs and 8.4 lbs?
2. What percentage of babies weigh less than 4.0 lbs?

2.6 Extra Practice

1. Compute the mean and median for the annual income dataset. Based on these values is the data skewed to thee right or left? Generate a histogram of the data to confirm your thoughts. Data

2. A survey recorded the time spent on homework in two different grades. The five-number summaries are recorded below. Compute the IQR for each grade. What are the main differences in the distributions?

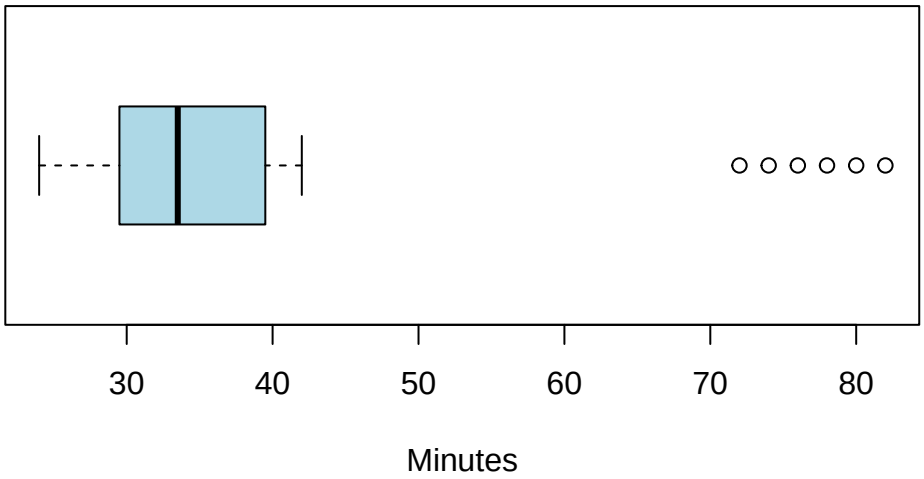
	Grade 9	Grade 11
Minimum	60	40
Q1	90	70
Median	110	120
Q3	130	200
Maximum	160	260

3. The principal tells you that, based on the above sumamary data, 9th graders and 11th graders spend a similar amount of time on homework. How might you respond to the principal?

4. The box plot shows commute times (in minutes) for 36 employees.

a) Estimate the median, Q1, and Q3. What is the IQR?

b) Would the mean be larger or smaller than the median? Why?



5. A news article reports: "The mean commute time is 25 minutes with a standard deviation of 40 minutes." What does this suggest about the shape of the distribution? Why might this statement be misleading?

6. Heights of adult women in the U.S. are approximately normal, with a mean of 64 inches and a standard deviation of 2.5 inches.

a) What percentage of women are between 61.5 and 66.5 inches tall?

b) What percentage of women are taller than 69 inches?

c) What percentage of women are between 59 and 66.5 inches tall?

3 Examining Relationships

In Topic 1 we looked at ways to visualize data about a single variable. Now we want to learn ways to examine the relationship between *two variables*.

When we look at two variable we are interested in how one impacts the other. The **response variable** or dependent variable is the variable that gives the outcome of the study. The **explanatory variable** is the variable that claims to impact the response variable.

Example 3.1. Identify what is the explanatory and response variable in the following study questions:

1. Does the amount of fertilizer affect how tall tomato plants grow?
2. Does a car's weight impact its mileage?
3. Do people who sleep more have lower stress levels?

> **!** Identifying Response and Explanatory
>
> The roles aren't always clear-cut: sometimes either variable could reasonably influence the other. In these cases, the study design is key in dictating which is the explanatory and which is the response variable.

3.1 Side-by-Side Boxplots (C $\rightarrow$ Q)

If the explanatory variable is categorical and the response variable is quantitative, then one way to visualize the potential differences in distribution is by using side-by-side boxplots.

Example 3.2. Below is a dataset of 189 births from a 1986 study at Baystate Medical Center (in Massachusetts). What similarities and differences do you notice between babies born from smokers versus nonsmokers? What can you not tell by the boxplots that you might be interested in knowing about the data?

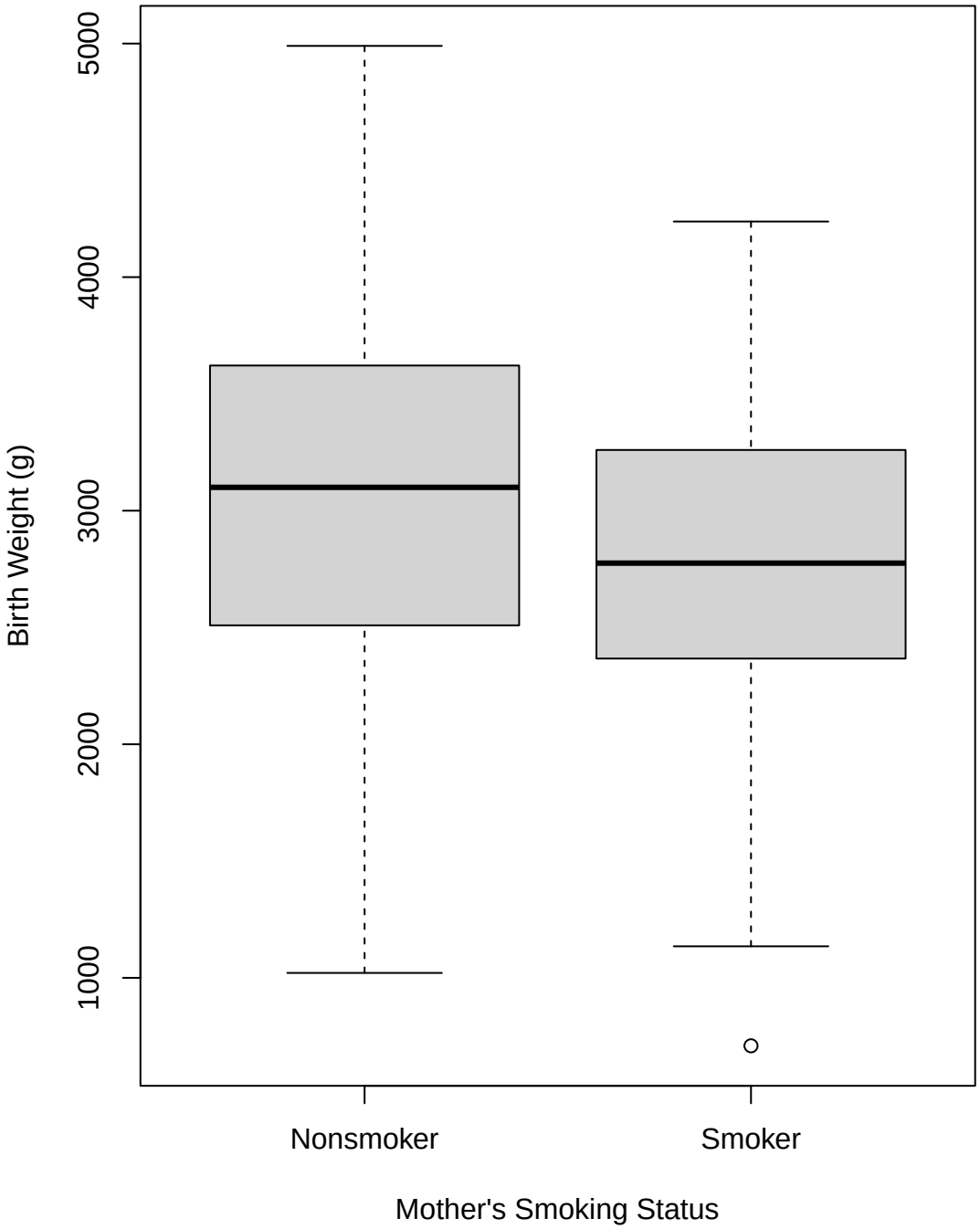


Figure 3.1: Side-by-side boxplots of birth weight based on mother's smoking status.

Answer

From the bokplot it looks like the average baby weight is higher for nonsmoking mothers, however there is a large overlap in the IQR. The lower end of the boxplots look similar. I would be interesting in learning how many participants were smokers/nonsmokers, which we cannot tell from the boxplots.

Association does not equal causation!

There may be a **lurking variable** which affects both the explanatory and response variable. This is why you should be careful about drawing conclusions just from statistical data.
Good study design should help eliminate potential lurking variables, however they are impossible to completely eliminate.

Example 3.3. What are potential lurking variables that could impact the study in Example Example 3.2?

Example 3.4. The above data is available for you to manipulate on StatCrunch here. Use the data to answer the following questions:

1. How many mothers were smokers/nonsmokers?
2. Is there a difference between the mother's weight for smokers/nonsmokers?
3. Is there a difference in the mother's age for smokers/nonsmokers?

Another method of visualizing relationships between a categorical explanatory variable and a quantitative response variable is by using a stacked histogram.

Here we see that the histogram is colored by player position. From this visualization we can see that defenders typically don't score much.

StatCrunch will let you do something similar, although it won't let you exactly stack the bars like above. In StatCrunch you can choose Graph > Histogram. Then you can select your response variable and group by your explanatory variable. Scroll all the way down to the multiple graph section and select the same X and Y axis. You can also use the rows and columns per page to change how many histograms are viewed at once.

3.2 Contingency Tables (C $\rightarrow$ C)

Example 3.5. A college polls 200 students to determine whether there is a relationship between a student's class year (freshman, sophomore, junior, or senior) and whether or not they live off campus. This data is shown in the **contingency table** below.

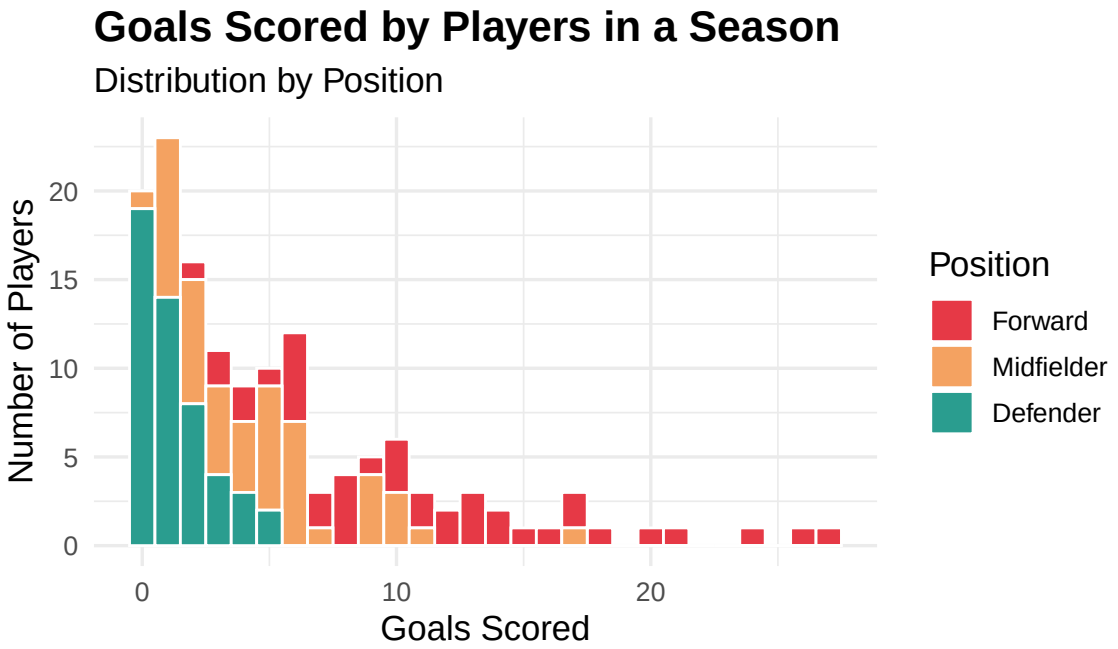


Figure 3.2: Stacked histogram of goals scored by different positions.

a) What percentage of freshmen live off campus?
b) What percentage of seniors live off campus?

Class Year	On Campus	Off Campus	Total
Freshman	42	13	55
Sophomore	35	15	50
Junior	18	30	48
Senior	9	38	47
Total	104	96	200

Answer

a) $(13/55) \times 100 = 23.6\%$
b) $(38/47) \times 100 = 80.9\%$

We can convert a contingency table to percents. To do this we convert counts to percentages when restricting to a single explanatory variable. For example the row

Class Year	On Campus	Off Campus	Total
Freshman	42	13	55

becomes

Class Year	On Campus	Off Campus	Total
Freshman	$(42/55) \times 100 = \mathbf{76.4}\%$	$(13/55) \times 100 = \mathbf{23.6}\%$	100%

In this case the rows should each add up to 100%. If our explanatory variables were in the columns, then we would make the columns sum to 100%.

Example 3.6. Using the data from the baby weight study above, create a contingency table on StatCrunch of the mother's smoking status and race. Data

1. For the mother's at the hospital, does race impact whether or not they smoke? Use data from the table to support your answer.
2. Does smoking impact whether the mother's are diagnosed with hypertension?

3.3 Scatterplots (Q -> Q)

Example 3.7. We want to determine if more time spent studying improves the score received on an exam. The amount of time spent studying was self reported by students. Open the data in StatCrunch.

Since we want to know if study time improves exam score, the explanatory variable is study time and the response variable is exam score. We can plot these using a scatter plot (Graph > Scatter Plot). We should set the study time as the x variable and the exam score as the y variable.

We see from the scatter plot that hours spent studying is **positively correlated** with exam scores. This means that the more hours spent studying, the higher of an exam score a student tends to receive. Notice that the points appear to show a "fuzzy" line with a positive slope.

Average hours of exercise weekly is **negatively correlated** with resting heart rate. This means the more hours exercised, the lower the resting heart rate.

We use the words **strong**, **moderate** and **weak** to describe how linearly correlated the data is. Below is an example of each,

Above the value r is the **correlation coefficient** for the data. Don't worry about the formula for computing r as we will use StatCrunch for this computation. See how to use it below,

Example 3.8 (Correlation Coefficient)**.** Open this data in StatCrunch. Make a scatter plot with longevity as the explanatory variable and gestation period as the response variable. Then compute the correlation coefficient of the data.

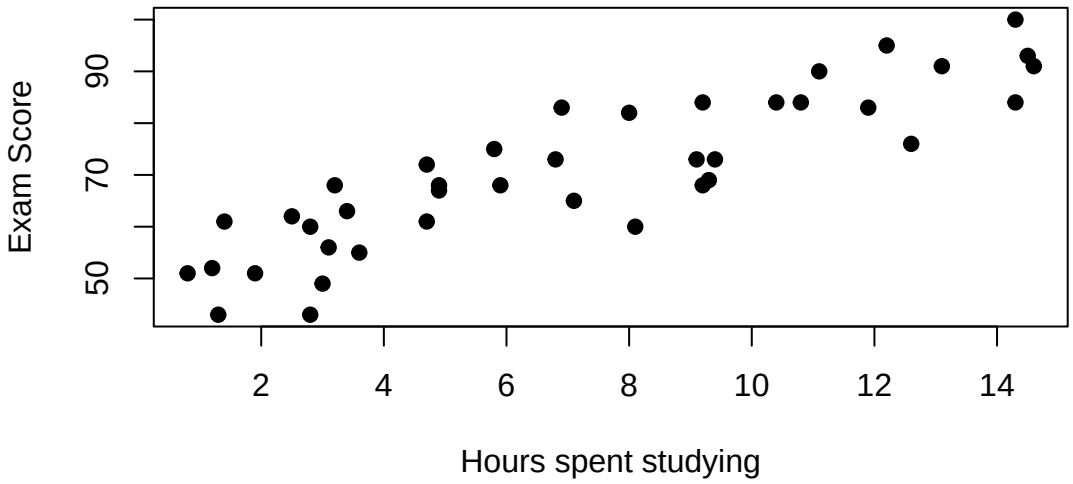


Figure 3.3: Scatter plot of time spent studying and exam scores.

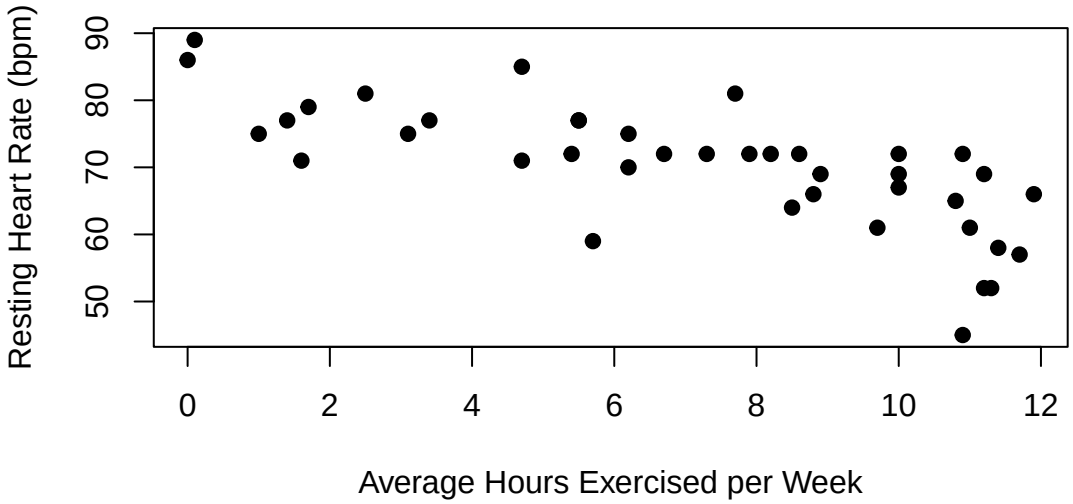


Figure 3.4: Scatter plot of weekly exercise hours and resting heart rate.

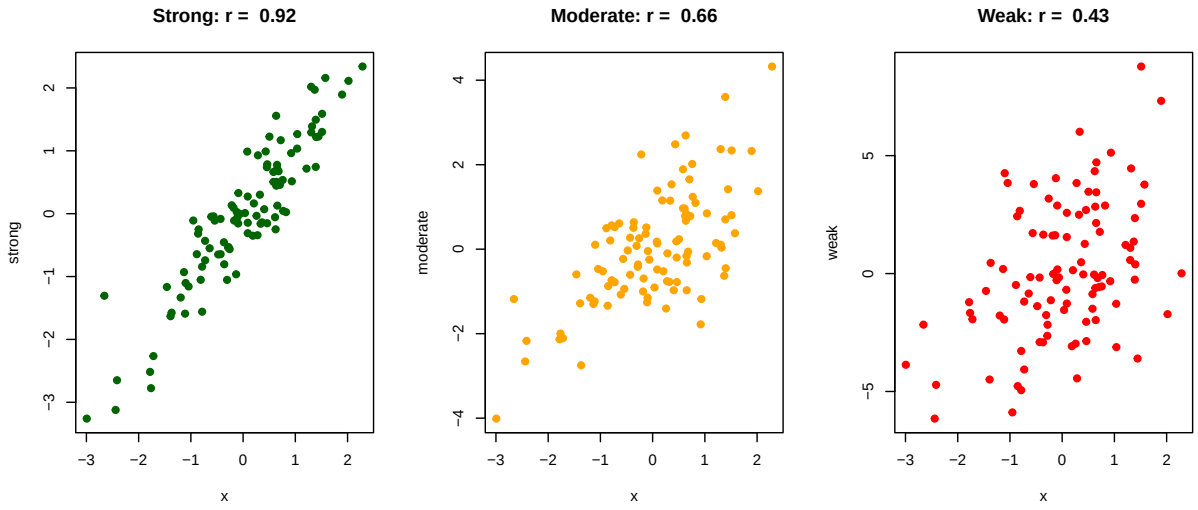


Figure 3.5: Strength of correlation.

The correlation coefficient r can measure the strength of the correlation, but only after we use the scatterplot to justify that it is linear! If our data is approximately linear, then the sign of the correlation coefficient tells us whether it is positive or negatively correlated. The magnitude of the correlation coefficient tells us the strength. There are a lot of different bench marks you will see for what r value is "moderate" versus "strong." You should think of these benchmarks as a guideline and that they are context dependent.

Plenty of data is not linearly correlated! Let's see some examples.

! Don't Over-Value r

The correlation coefficient should only be used as *supporting evidence* alongside the scatter plot. When a scatter plot looks linear, then r can tell us the strength and direction of that linearity. We cannot use r alone to justify linearity!!

3.4 Linear Regression

If we do feel a line is linear we can perform **least-squares linear regression**. This creates a line that best fits the data. We will not do this by hand, we will use StatCrunch. It is important to be able to interpret the StatCrunch output though.

Example 3.9. Open this data in StatCrunch. Perform the linear regression on the data. What is the equation for the line of best fit? Write a sentence to interpret the slope in the context of the explanatory and response variables.

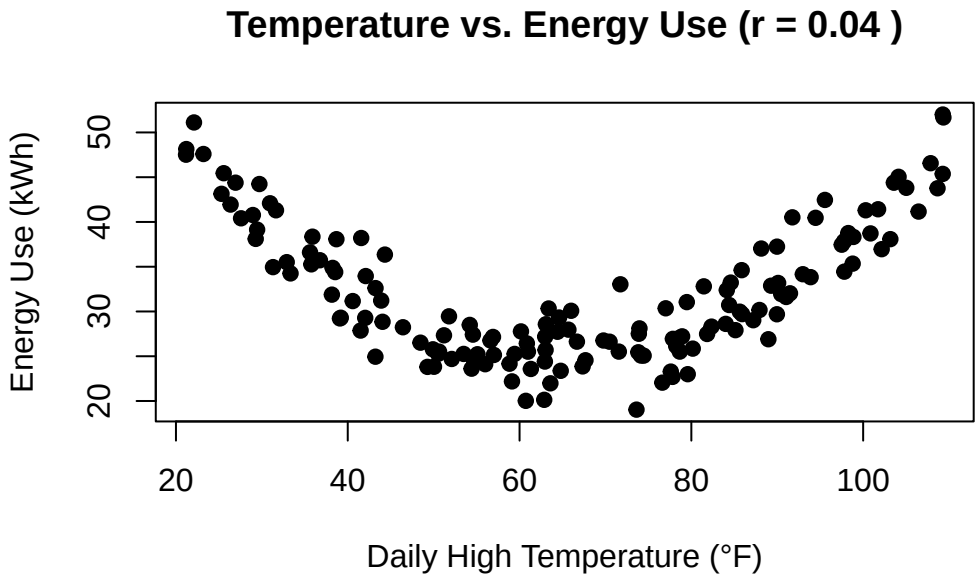


Figure 3.6: Daily home energy use is highest on cold and hot days and lowest at mild temperatures.

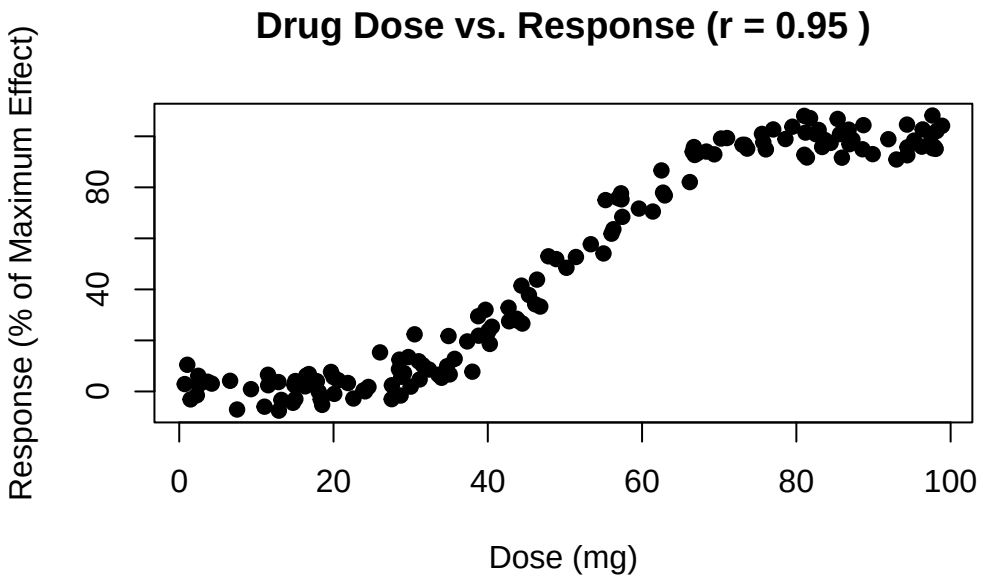


Figure 3.7: Response stays low at small doses, rises steeply, then plateaus near the maximum effect.

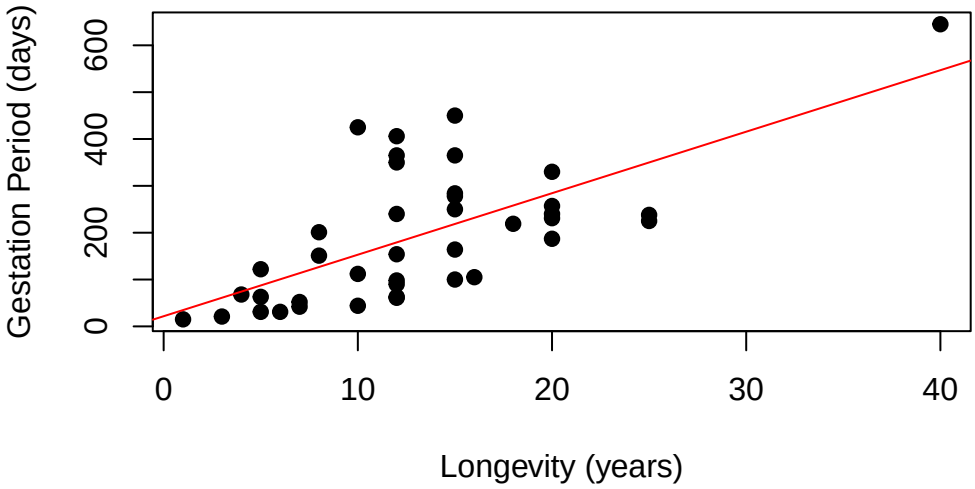


Figure 3.8: Longevity and gestation period for a variety of animals.

Answer

Under the table **Parameter Estimates** we want to look at the **intercept** and **slope** estimate values to generate the line of best fit.
Intercept: 21.71
Slope: 13.13
Recall that longevity is our response variable–this means for our line it is the *input* (x variable). Gestation will be the *output* (y variable). So our line has the form,

$$\text{gestation} = 13.13 \times (\text{longevity}) + 21.71$$

The slope tells us that, on average, each year of longevity amounts to an additional 13.13 days of gestation. The intercept tells us for a animal with a longevity of 0 years, the gestation period is, on average, 21.71 days. This is an impossible scenario though, so it is outside of our domain of interest.

Example 3.10. Use the line of best fit computed in Example Example 3.9 to approximate the gestation period of a North American river otter, which has a typical lifespan of 12 years.

Answer

Using our line, we estimate that the North American river otter will have a gestation period of $13.13 \times 12 + 21.71 = 179.27$ months.
According the the internet, the river otters have a gestation period of 285-375 days due to a process called delayed implantation. The more you know, eh?!

! Estimation

You should avoid using linear regression to estimate values that fall outside of the range of your data. When we zoom in on functions they almost always look linear, but zooming out can reveal more complicated behavior. Just because a line fits the data you have, doesn't mean it will still look linear when we "zoom out."
For example, the line in Example Example 3.9 should *not* be used to approximate the gestation time of an ORca, which averages 90 years of life. Plugging this in gives an approximate gestation period of 1,203 days or ~40 months–much longer than their actual gestation period of 15-18 months.

Example 3.11. Data. Using the provided dataset, answer the following:

1. Describe the relationship in words between the car's weight and its miles per gallon. Use the weight as the explanatory variable.
2. Compute the correlation coefficient.
3. Use linear regression to write a line that estimates the car's mpg from it weight.
4. Estimate the mpg of a car that weighs 4250 lbs.

4 Sampling and Study Design

4.1 Sampling

Sampling is the way we select participants for a study. Sampling design is *crucial* for creating an unbiased study.

Simple random sampling involves selecting randomly from the entire population. In practice, there are many limiting factors involved in generating a truly simple random sample.

Example 4.1 (Simple Random Sample)**.** What is the *intended* population of each study? How do the following random samples fail to reflect the entire population or be genuinely random?

1. A city wants to know how adult residents feel about expanding bus service. Planners get the list of every account on the city's water and electric billing system and randomly select 1,000 accounts to survey.
2. A polling firm wants to estimate support for a ballot measure among adults in Ohio. It buys a list of all published residential phone numbers in the state and uses software to randomly select 1,500 numbers to call.
3. A grocery store chain wants to know how satisfied its customers are with checkout wait times. It pulls the list of all transactions from the past month in its system and randomly selects 500 of them. It then contacts each matched customer through the email linked to their loyalty account.

Answer

1. They are only sampling from names listed on a utility bill. Those who live with roommate ro cooperative housing are excluded from the survey.
2. This survey is excluding households without a landline, which could lead to bias in the sample.
3. There are two main issues with this. The survey is not truly random because customers who shop more are more likely to get contacted. Additionally, only those signed up for rewards are contacted.

We refer to the **sampling frame** as the actual population where the sample was taken from. Like in the above example, the sampling frame can be different from the intended population.

Systematic sampling is when we choose from the population in a systematic fashion, e.g. picking every 5th name in an alphabetized list, picking every 3rd person to show up at a clinic.

Cluster sampling is when we break the population down into clusters, randomly select some of the clusters, and use every individual in the selected clusters.

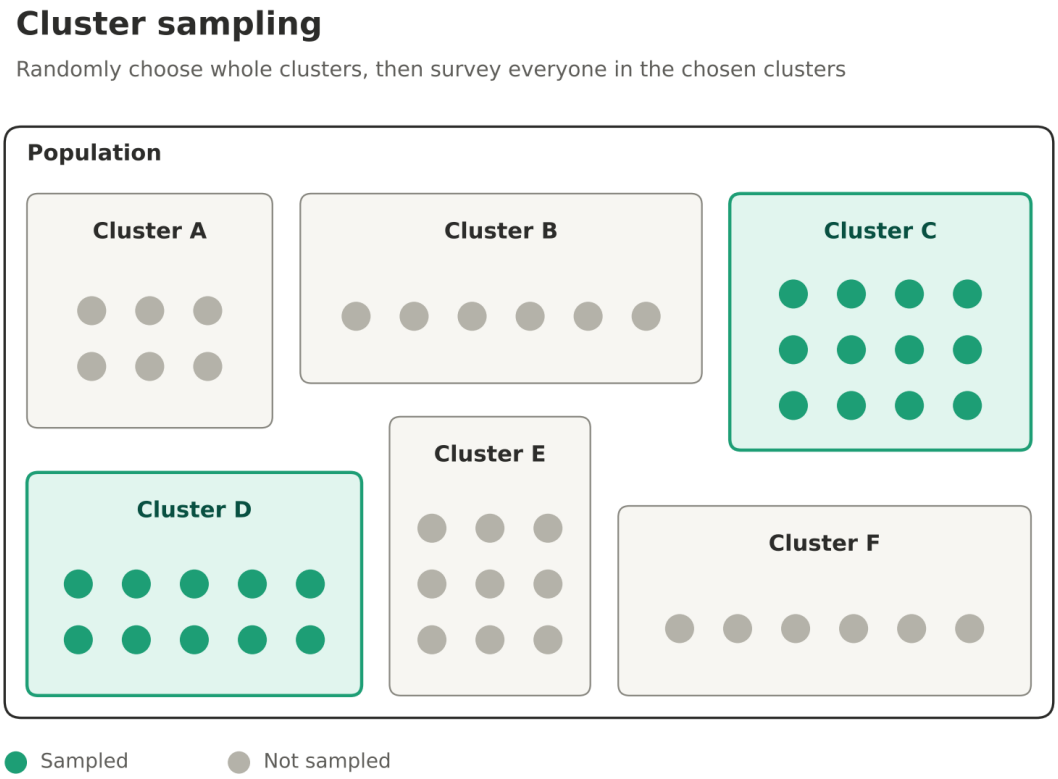


Figure 4.1: Cluster Sampling. Image generated by Claude.

Example 4.2. When it is a bad idea to use cluster sampling?

Answer

Don't use cluster sampling if your clusters vary widely in make up (with relation to your survey). For example, suppose I was asking the question "Does your household have access to a car?" Clustering by cities in the Sound are would be a bad idea, as we expect

those in Seattle & Tacoma to have lower rates of car access than a Bainbridge Island or Gig Harbor.

Stratified sampling is when we break down a population into strata and sample randomly *from each* strata. This is a good type of sampling to do if you can identify potential groupings in the population where you expect major differences.

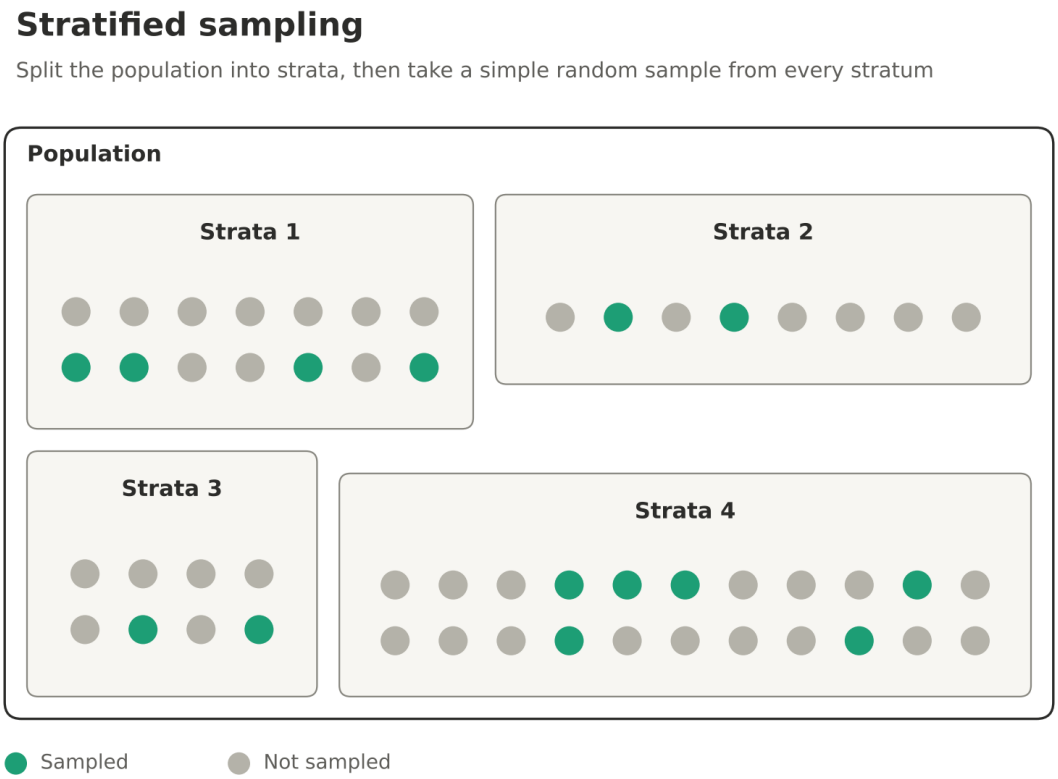


Figure 4.2: Stratified Sampling. Image Generated by Claude.

Example 4.3 (Stratified Sampling)**.** The county transit authority wants to estimate the percentage of Cuyahoga County (Cleveland, OH and surrounding area) households that don't have access to a car, to help plan bus routes. They decided to use *stratified sampling* to pick a random sample of the county population. For each city/town in the county they sample roughly 15% of the population.

! Clusters Versus Strata

There's no difference in the meaning between the terms *cluster* and *strata*. In both cases we are breaking the population down into groups.

Multistage sampling is when a mix of methods is applied to different levels.

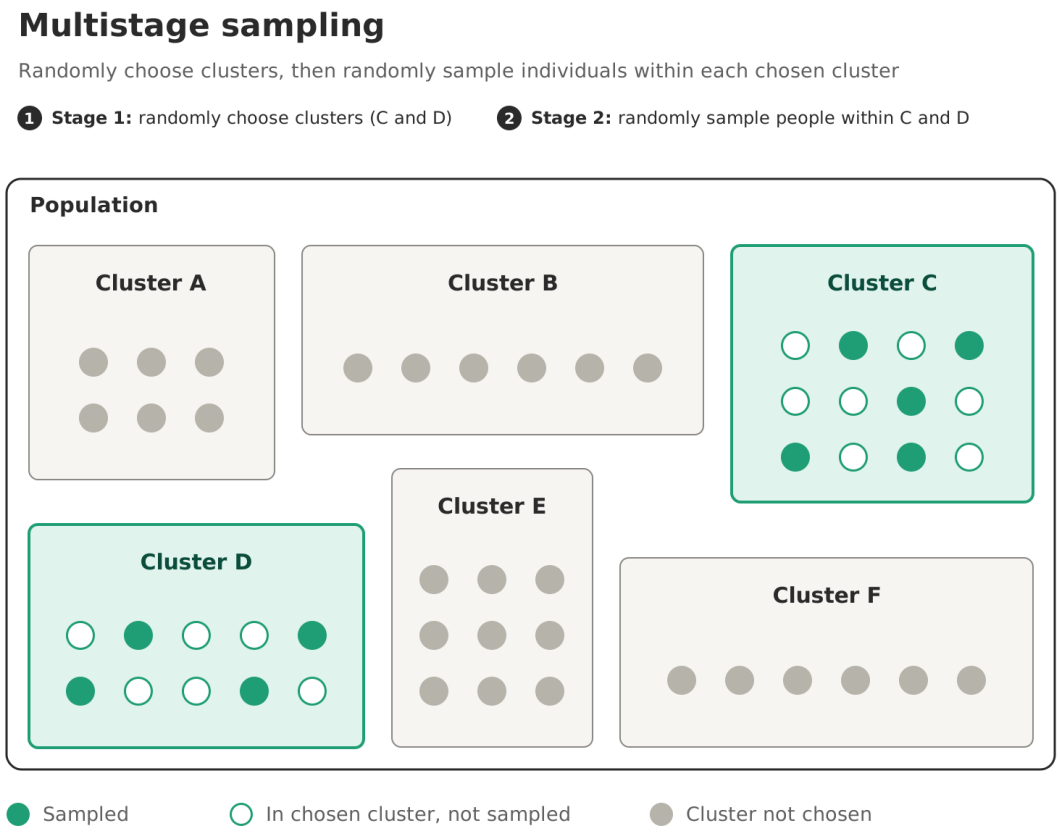


Figure 4.3: An example of multistage sampling. Image generated with Claude.

Example 4.4 (Identify the Sampling)**.** For each scenario identify the type of sampling used as *simple*, *systematic*, *cluster*, or *stratified*.

1. A teacher of a 500 person lecture wants to know how much time students on average on each homework assignment. The class contains freshmen, sophomores, and juniors, so the teach randomly selects 50 students from each class to sample.

2. A grocery store manager wants to estimate customer satisfaction. Starting with a randomly chosen customer among the first 20 to check out on Saturday, the manager surveys every 20th customer who passes through the checkout line for the rest of the day.
3. A school district wants to know how many hours per week high school students spend on extracurricular activities. The district has 40 homerooms, so it randomly selects 6 homerooms and surveys every student in those homerooms.

Answer

1. This is stratified sampling, since we choose a random sample from each class.
2. This is systematic sampling.
3. This is cluster sampling, since we survey all the student in each homeroom.

4.2 Study Design

There are three types of study design we will focus on learning:

- An **observational study** is when participants are asked about their variables in their life *without* researcher interference. An observational study is **prospective** if variables are recorded as they happen. An observational study is **retrospective** if participants recall information about the past.
- A **survey** is where where participants are asked to report on certain variables related to themselves, typically through a form, survey, or interview.
- An **experiment** is when researchers interfere instead of measuring variables as they occur naturally. Researchers assign the explanatory variable to participants and see how it impacts the response variables.

Surveys

A survey *is* a type observational study, however it earns a distinction because surveys ask participants to express their feelings/opinions. An observational study (not survey), typically relies on observation, tracking, or medical records.

Example 4.5 (Study Types)**.** Determine whether each study is an observational study, survey, or experiment.

1. A researcher records how many cars run a red light at a busy intersection each day.
2. Students are randomly assigned to study with or without music, then take the same quiz.

3. A school emails parents a questionnaire asking how satisfied they are with the lunch menu.
4. Doctors review hospital records to compare heart disease rates in smokers and non-smokers.
5. Tomato plants are randomly given one of three fertilizers, and their yields are measured.
6. A pollster calls 1,000 voters and asks which candidate they plan to support.

Answers

1. observational study (prospective)
2. experiment
3. survey
4. observational study (retrospective)
5. experiment
6. survey

4.3 Causation and Experiments

A *good* experiment should use randomization to control for other factors potentially impacting the response variable.

Consider an experiment where we trying to measure the efficacy of a new weight loss drug. The explanatory variable in this case is the new drug and the response variable would be measures of weight loss.

In the experiment split individuals into a**treatment group**, who receive the drug and a **control group** who receive do not receive the drug. These groups should be *randomly assigned* and we should do so in a way that makes the two groups similar with respect to other factors that could impact weight loss.

Example 4.6. What are other factors we might want to control for when randomly assigning our groups?

Another good idea would be to make the experiment **blind** this means the participants should not know whether they are in the treatment or control group as it could impact their behavior. We could administer a **placebo** to the control group.

We could go even further and make the experiment **double-blind** meaning that even the researchers don't know what group the participant was assigned to.

Example 4.7. Give an example of an experiment that is impossible to make blind. Give an example of an experiment that could be blind, but not double-blind.

4.4 Survey Questions

Survey questions should be carefully thought through in order to be effective. When drafting questions, one should:

- Think about whether a question is better *open* or *closed*,
- Avoid planting ideas or leading questions,
- Avoid complicated questions addressing more than one idea,
- Avoid unbalanced response options.

Example 4.8 (Biased Survey Questions)**.** Determine whether each survey question is biased or unbiased. If it is biased, explain why.

1. Do you agree that the school day starts too early?
2. How many hours of sleep did you get last night?
3. Should the city waste more taxpayer money on a new stadium?

Example 4.9 (Choosing the Better Question)**.** For each pair, decide which survey question is better and explain why.

1. a. Do you not disagree that homework should not be eliminated?
   b. Do you think homework should be eliminated?
2. a. To what extent do you concur that the implementation of later start times would positively influence your academic performance and overall well-being?
   b. Would a later school start time help you do better in school?
3. a. How satisfied are you with the school library?
      Somewhat satisfied Satisfied Extremely satisfied Dissatisfied
   b. How satisfied are you with the school library?
      Very dissatisfied Dissatisfied Neutral Satisfied Very satisfied

5 Basics of Probability

The **probability** of an event is the likelihood of the event occurring. If we denote an event by E then we write

$$P(E)$$

to denote the probability that event E happens. Probabilities should be numbers between 0 and 1.

$$E \text{ is impossible} \implies P(E) = 0.$$
$$E \text{ always occurs} \implies P(E) = 1.$$

> **!** Zero Probability
>
> The *converse* of the above statements are not true. Meaning that if an event has probability 0 it does not mean it is impossible. This only happens in cases of infinite *sample space* though, so don't worry about it. :)

An event has a **theoretical** probability, where we infer the probability from the nature of the situation.

Example 5.1. If I flip a **fair coin** there probability it lands on heads is 0.5.

If I roll a fair **6-sided** dice, the probability it lands on any of the numbers is $1/6 \approx 0.1667$.

These are a **theoretical probability**.

We can also use data to conjecture probabilities. We will call this an **empirical probability**.

Example 5.2. Flip a (fair) coin 30 times and record the number of times it lands on heads. Dividing this number by 30 will give an empirical probability.

According to the **Law of Large Numbers** the empirical probability should be close to the theoretical probability if the number of random experiments is large.

The coin flip example is silly because the theoretical probability is easily attainable. Empirical probability are useful when the actually theoretical probabilities are not attainable.

Example 5.3. Suppose I want to know the probability of a penalty kick being scored amongst international mens soccer teams. A theoretical probability would be impossible to compute here, but I could use historical data, like the data found here, to get an empirical probability.

Example 5.4 (Empirical Probability)**.** A school surveyed 200 randomly selected students about how they usually get to school. The results are shown below.

Transportation	Number of Students
Car	84
Bus	56
Walk	36
Bike	16
Other	8
Total	**200**

Suppose a student from this school is selected at random. Use the data to find each probability.

1. The student rides the bus.
2. The student bikes to school.
3. The student walks or bikes to school.
4. The student does **not** get to school by car.
5. If the school has 1,500 students, about how many would you expect to ride the bus?

Answer

1. $56/200 = 0.28$
2. $16/200 = 0.08$
3. $52/200 = 0.26$
4. $116/200 = 0.58$
5. $0.28 \times 1500 = 420$.

5.1 Random Experiments, Sample Space, and Outcomes

A **random experiment** is a process that can be repeated under the same conditions, where all possible outcomes are known in advance but the outcome of any single trial cannot be predicted. Examples include,

- Flipping a coin X amount of times.
- Choosing a random person and checking their eye color.
- Rolling two dice until you get a pair.

An **outcome** is any possible result of the experiment. The **sample space** of a random experiment is the set of all possible outcomes.

Example 5.5. Todd rolls a dice and records the outcome. The sample space is,

$$\{1, 2, 3, 4, 5, 6\}$$

Example 5.6. Julia flips a coin twice and records the results. The sample space is,

$$\{HH, TT, HT, TH\}$$

Notice that TH and HT are two different outcome–order matters in this experiment.

Example 5.7. Dr. Gordo randomly selects a patient file and checks their blood type. The sample space is,

$$\{O, A, B, AB\}$$

Example 5.8. Julia flips a coin three times and records the results. What is the sample space?

> Tip
>
> HHH, TTT, HTT, THH, HHT, TTH, HTH, THT.

With our definition of sample space, we can redefine an **event** as a subset of the sample space.

Example 5.9. Julia flips a coin three times. Let A be the event that there is at least two tails flipped. Which outcomes from the sample space above are included in this event?

Tip

TTT, HTT, TTH, THT.

Example 5.10. Margot rolls two 6-sided dice and records the outcomes. What is the sample space?

Let A be the event that the sum of the two rolls is at least 8. What outcomes are included in this event?

Tip

The sample space is: 11, 12, 13, 14, 15, 16, 21, 22, 23, 24, 25, 26, 31, 32, 33, 34, 35, 36, 41, 42, 43, 44, 45, 46, 51, 52, 53, 54, 55, 56, 61, 62, 63, 64, 65, 66.
The outcomes where the sum is at least 8 are: 26, 35, 36, 44, 45, 46, 53, 54, 55, 56, 62, 63, 64, 65, 66.

5.2 Naive Probability

If all outcomes in the sample space are equally likely, then the problem of an event is

$$P(E) = \frac{\text{number of outcomes in the event}}{\text{size of the sample space}}.$$

Example 5.11. What is the probability of flipping a coin three times and getting exactly 1 head?

Answer

From earlier we know the sample space has 8 events. All of these events are equally likely. The outcomes with exactly one head are:: HTT, THT, TTH. So the probability is $3/8 = 0.375$.

Example 5.12. A coin is flip 3 times or until a head is flipped. What is the sample space? Can I use the naive probability definition to compute the probability of flipping at least 1 tail? Explain.

Tip

The sample space is,

$$\{H, TH, TTH\}.$$

I cannot use the naive definition because the three outcomes are *not* equally likely.

Example 5.13. Two 6-sided dice are rolled. What is the probability that the sum of the two numbers is less than 6?

Answer

The sample space is the same as in the solution to Example Example 5.10. All of these events are equally likely, so we can compute the probability by denoting how many outcomes have sum less than 6:

$$11,\ 12,\ 13,\ 14,\ 21,\ 22,\ 23,\ 31,\ 32,\ 41.$$

So the probability is $10/36 = 0.278$.

5.3 Rules of Probability

The **complement** of an event A is all outcomes where A does not happen.

Example 5.14 (Complement)**.** Suppose I roll a 6-sided dice. Let A be the event that I roll a number divisible by 3. Therefore

$$P(A) = \frac{2}{6} \approx 0.33.$$

The complement of A, denoted A^c is the event that I roll a number *not* divisible by three, e.g. 1, 2, 4, or 5. Thus,

$$P(A^c) = \frac{4}{6} \approx 0.67.$$

For an event and its complement, the following relationship holds

$$\boxed{P(A) = 1 - P(A^c)}$$

Example 5.15. State the complement of the following events. You do not need to compute any probabilities.

1. The event that a randomly selected patient has blood type O.
2. I roll two 6-sided dice. Let A be the event that the sum of the roll is at least 10.
3. A flip a coin until I flip heads. Let A be the event that I get heads by the 10th flip.
4. I roll two 6-sided dice. Let A be the event that neither roll is a 6.

In some instances, it is easier to compute the probability of the complement. Thus we can use the boxed relationship to speed up the problem.

Example 5.16. I roll two 6-sided dice. Compute the probability that the sum of the two dice is at least 3.

Answer

We know that there are 36 outcomes when rolling a dice twice. Three of those outcomes sum to *less than* 3: 11, 12, 21. So the probability of rolling *at least* a 3 is,

$$1 - \frac{3}{36} = \frac{33}{36} \approx 0.92.$$

Let A and B be events. The **union** of A and B, $A \cup B$, is the event where at least one of the happens. For example, suppose I randomly select a person on the street and let

$$A = \text{the person has a dog}, \ B = \text{the person has a cat}.$$

Then the union is

$$A \cup B = \text{the person has a dog } \textbf{or} \text{ a cat}.$$

The **intersection** of A and B, $A \cap B$, is the event where they both happen. So in the example above,

$$A \cup B = \text{the person has a dog } \textbf{and} \text{ a cat}.$$

Two events are **disjoint** if it is impossible to satisfy both at the same time. Mathematically, they are disjoint if $A \cap B$ is impossible (e.g. contains no outcomes).

Example 5.17. Determine whether or not the following events are disjoint.

1. Roll a dice. $A =$ the dice shows an even number, $B =$ the dice shows an odd number.

2. Roll a dice. A = the dice shows an even number, B = the dice shows a number greater than 3.
3. Flip a coin twice. A = both flips are heads, B = at least one flip is tails.
4. Roll two dice. A = the sum is 8, B = both dice show the same number.
5. Randomly select a high school student. A = the student is a freshman, B = the student is a senior.
6. Randomly select a high school student. A = the student plays a sport, B = the student is in band.

Answer

1. Disjoint, a number cannot be both even and odd.
2. Not disjoint, for example a roll of 4 satisfies both.
3. Disjoint.
4. Not disjoint.
5. Disjoint.
6. Not disjoint.

If events A and B are disjoint, we can add their probabilities to compute the probability of the union

$$P(A \cup B) = P(A) + P(B)$$

But if two events are not disjoint, we should use the principle of **inclusion-exclusion**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Example 5.18. A high school has 400 students. Of these, 180 play a sport, 90 are in band, and 40 do both. A student is selected at random.

1. What is the probability the student plays a sport?
2. What is the probability that a student is in the band?
3. What is the probability that a student plays a sport and is in the band?
4. What is the probability that a student is in the band or plays a sport?
5. What is the probability a student does neither?

Answer

1. $180/400 = 0.45$.
2. $90/400 = 0.225$
3. $40/400 = 0.1$
4. Let S = plays a sport and B = in the band. Use inclusion-exclusion,

$$P(S \cup B) = 0.45 + 0.225 - 0.1 = 0.165.$$

5. Doing neither is the complement of being in band or playing a sport,

$$P((S \cup B)^c) = 1 - P(S \cup B) = 1 - 0.165 = 0.835.$$

Two events are **independent** if one event happening does not change the probability of the other event happening.

Example 5.19. Consider the events:

- It rains today.
- It is cloudy today.

These events are not independent, because if we know it is cloudy this increases the chance that it is going to rain. We say these events are **dependent**.

Example 5.20. Consider the events:

- A couples first baby is a girl.
- A couples second baby is a girl.

These events are independent, because the sex of the first child has no impact on the sex of the second child.

Example 5.21. Determine whether the following events are independent:

1. A coin is flipped twice. A: the first flip is heads. B: the second flip is heads.
2. A coin is flipped twice. A: the first flip is heads. B: at least one flip is heads.
3. A dice is rolled twice. A: the first roll is even. B: the sum of the two rolls is 7.
4. A dice is rolled twice. A: the first roll is a 6. B: the sum of the two rolls is at least 10.
5. A: a patient has blood type O. B: their child has blood type O.
6. A: the Seattle Mariners win. B: the average on the exam is more than 70%.

6 Random Variables and Probability Distributions

6.1 Random Variables

A **random variable** is an assignment of a numerical value to each possible outcome. We can then compute the probability of a random variable being equal to some value, or falling in a range of some values.

Example 6.1. Consider the random experiment when a flip a coin three times. Let X be the random variable that counts the number of heads that appear. There are four possible values X could be: 0, 1, 2, or 3.

The above is an example of a **discrete** random variable. If I can list out all the possible values of the random variable, then it is discrete.

A random variable is a **continuous** random variable if the possible values include any number in some interval.

Example 6.2. I randomly pick up a pile of sand at the beach. How much that piles weighs is a *continuous random variable*, since it can take a range of numbers.

In general, discrete random variables are things we count, while continuous random variables are things we measure. some examples of discrete random variable are:

- The number of coin flips until I flip a heads.
- The number of shots in a soccer game.
- The number of cars caught speeding at an intersection.
- The number of students who receive an A on an exam.

The following are continuous random variables:

- The weights of wrestlers.
- The height of skyscrapers.
- The temperature at the top of Mount Washington.

Note that sometimes the above can appear discrete, because we often round values. However, they are in fact continuous. The following variables are technically discrete, but they are frequently approximated as continuous:

- SAT, or other exam, scores.
- Annual salary of employees.

We will discuss later when approximation is useful and appropriate.

6.2 Discrete Probability Distributions

If we have a discrete random variable, we can describe the probability of each possible outcome with a **probability mass function**, also known as a **probability distribution**.

Example 6.3. Consider the random experiment when a flip a coin three times. Let X be the random variable that counts the number of heads that appear. We can write down the sample space to determine the probability of each possible value for X, since each outcome in the sample space is equally likely. Sample space:

$$HHH, TTT, HTH, THT, HTT, THH, HHT, TTH.$$

Now we can count the probability for each possible value of X.

x	$P(X = x)$
0	1/8
1	3/8
2	3/8
3	1/8

We read this as, $P(X = 0) = 1/8$, $P(X = 1) = 3/8$, etc.

We use capital letters, like X, to represent random variables and we use lowercase letters, like x, to represent placeholders for potential values.

Example 6.4. A random variable Y has the following probability distribution,

y	$P(Y = y)$
1	0.1
2	0.3

y	$P(Y = y)$
3	0.3
4	0.2
5	0.1

What is the probability that $Y = 2$? What is the probability that $Y \geq 3$?

Answer

From reading the table we see that,

$$P(Y = 2) = 0.3.$$

Note the the probability that $Y \geq 3$ is the same as the probability that Y is 3, 4, or 5. So,

$$P(Y \geq 3) = P(Y = 3) + P(Y = 4) + P(Y = 5) = 0.3 + 0.2 + 0.1 = 0.6.$$

A probability distribution should satisfy the following conditions:

1. For any x, $0 \leq P(X = x) \leq 1$.
2. When we sum over all possible values, the probability must equal 1.

Example 6.5. Does the following table represent a valid probability distribution?

x	$P(X = x)$
1	0.1
2	0.3
3	0.7

Answer

No. When we add up all the probabilities, the sum is 1.1 not 1.

Example 6.6. Does the following table represent a valid probability distribution?

x	$P(X = x)$
-2	0.25
-1	0.2
0	0.05

x	$P(X = x)$
1	0.3
2	0.2

Answer

Yes!

Example 6.7. Does the following table represent a valid probability distribution?

x	$P(X = x)$
1	0.3
2	0.3
3	0.3
4	0.3
5	-0.2

Answer

No, we cannot have negative probabilities.

6.3 Continuous Probability Distributions

To understand continuous distributions, it will help to start with a histogram. The following is a histogram of 1,000 adult female heights,

Example 6.8. What is the probability of selecting a random person with a height of at least 170 cm?

Answer

$(89 + 16 + 5 + 1)/1000 = 111/1000 = 0.111$

We divide by 1000 to get the correct probability. Instead of doing this, what if we rescaled the graph so the probability for each bar was equal to its area. Since each bar has **width** 5, we need to divide the heights by 200. This is known as a **density histogram**. Above the bars below I am showing the **area** of each bar not the height.

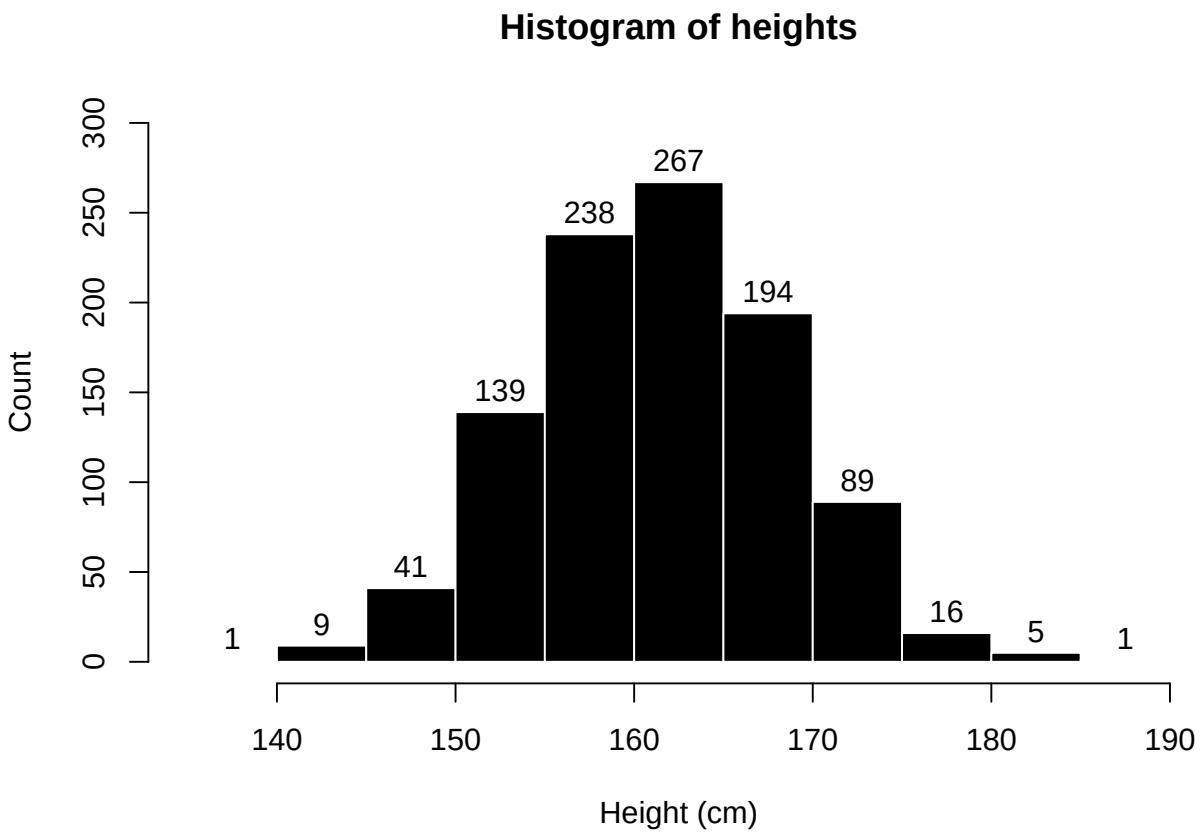


Figure 6.1: Histogram of 100 female heights.

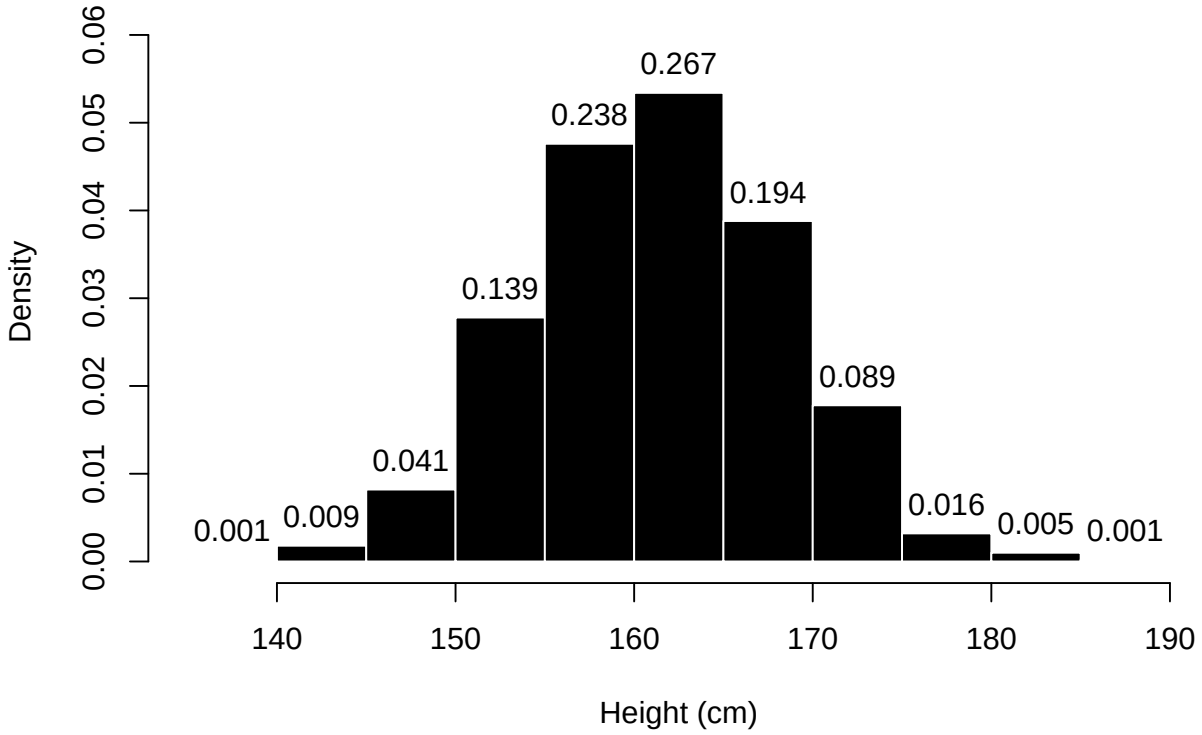


Figure 6.2: Density Histogram of 100 female heights.

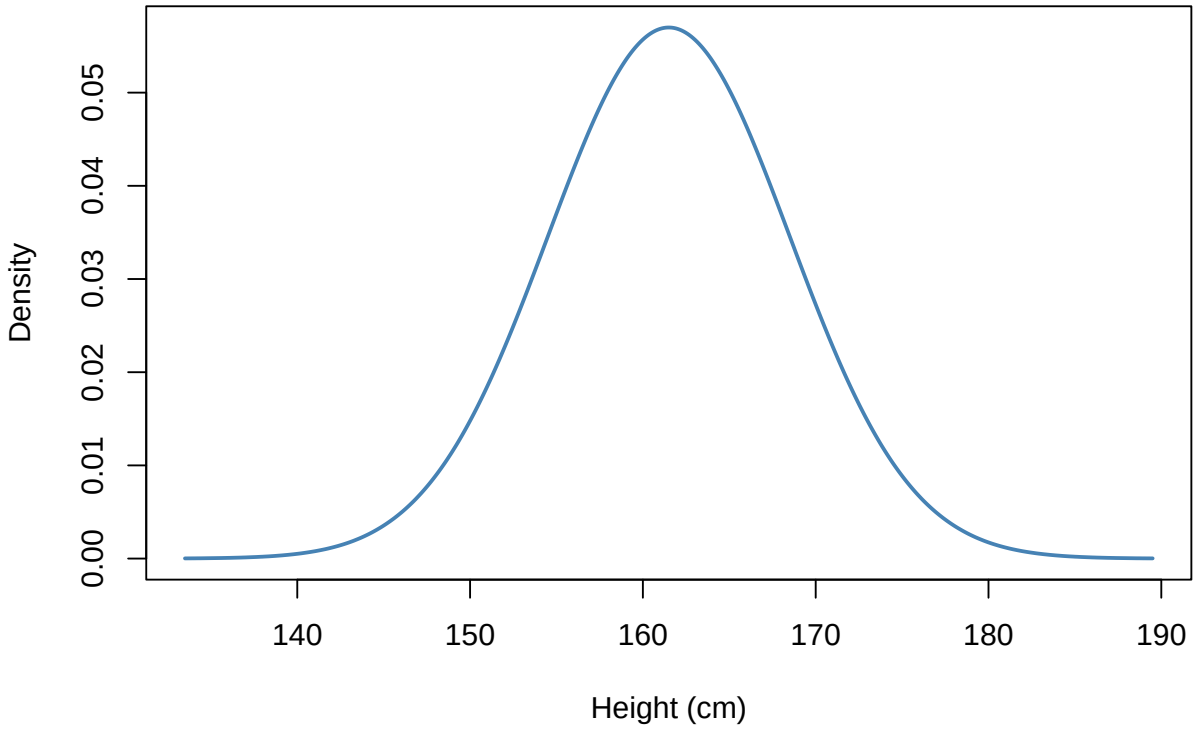


Figure 6.3: Distribution plot for female heights.

When a random variable is continuous we extend this idea to a continuous function known as a **probability density function**. The following density curve represents the heights,

The above is the probability density function of the **Normal distribution** with mean 161.5 and standard deviation 7. We have seen this curve before, when we learned the Empirical rule.

Again, if I wanted to know the probability that a randomly selected female has a height of at least 170 cm I would compute the area under the density curve.

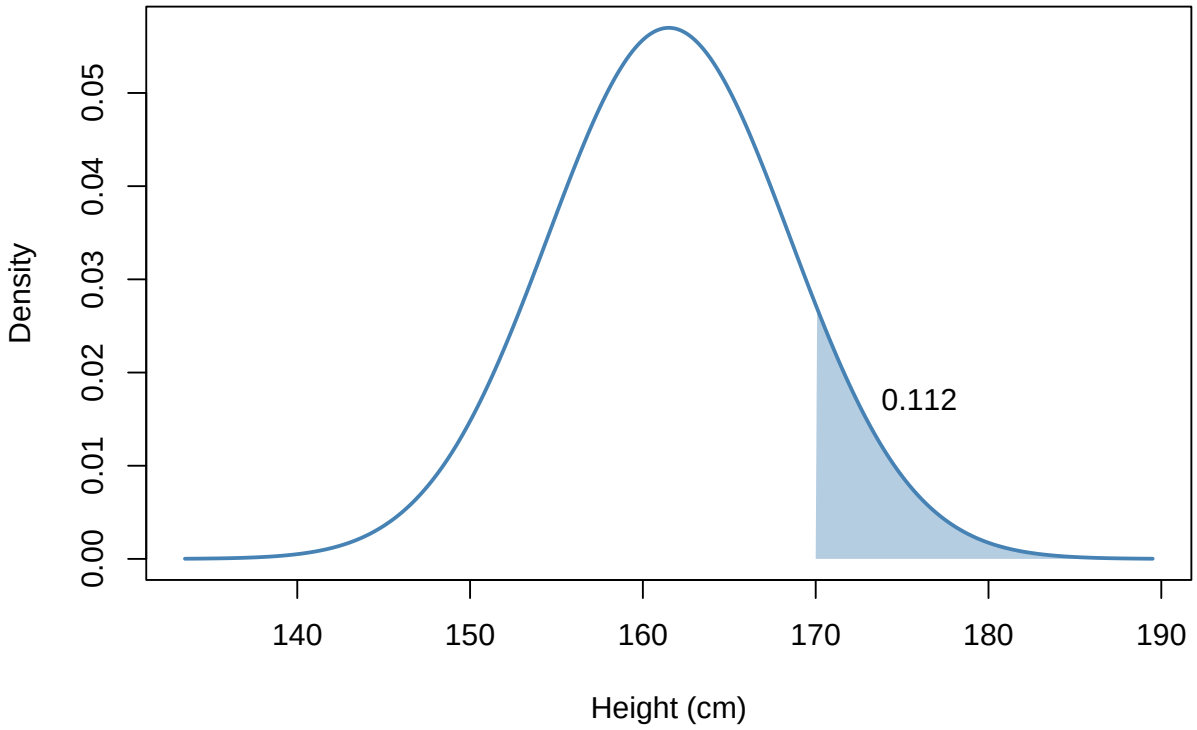


Figure 6.4: Distribution plot for female heights. The area of the shaded region is the probability of a random female having a height of at least 170 cm.

Similarly, if I wanted to compute the probability that someone has a height between 145 cm and 162 cm I would compute the following area.

To compute the actual area, we need to set up an integral. For us, we can use the **Normal Calculator** in StatCrunch. This will be our go to method for computing normal probabilities.

For a discrete random variable, whether or not the inequality is strict makes a difference.

Example 6.9. Roll two dice and let X denote the sum of the rolls. What is $P(X > 10)$? What is $P(X \geq 10)$?

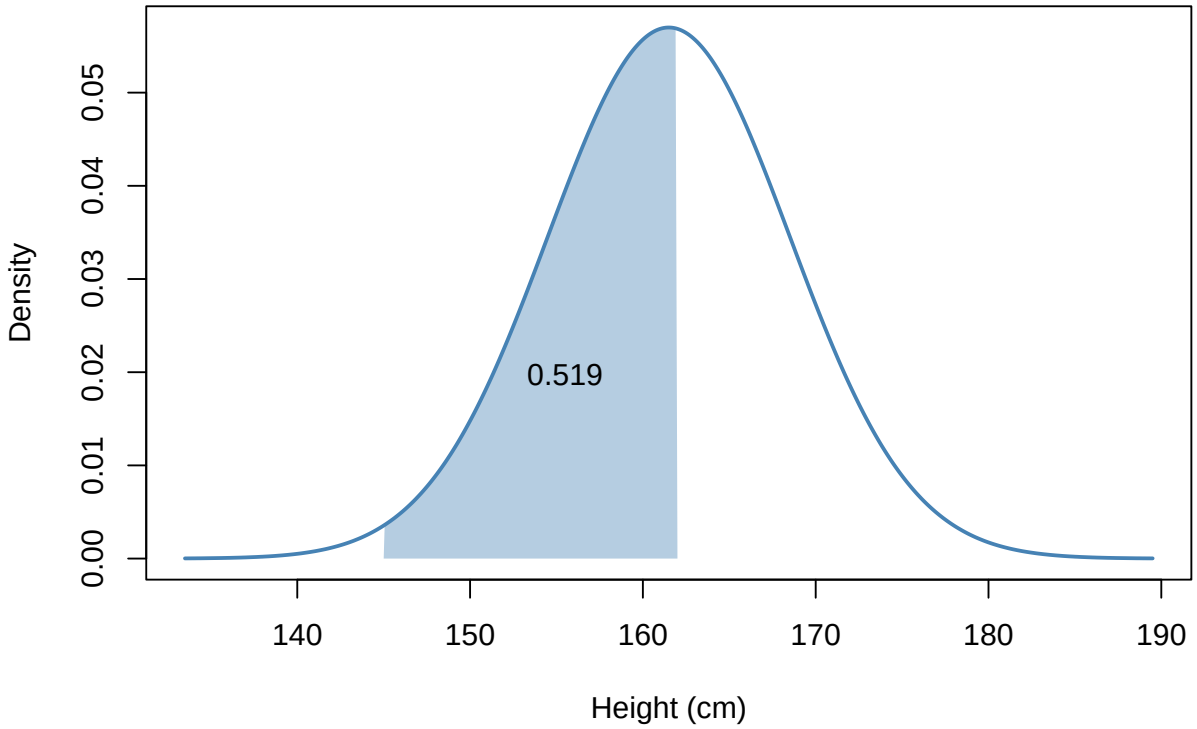


Figure 6.5: Distribution plot for female heights.

Answer

The sample space is 36 outcomes that are all equally likely. There is 1 outcome that sums to 12 (66), 2 outcomes that sum to 11 (56, 65), and 3 outcomes that sum to 10 (46, 55, 64). So,

$$P(X > 10) = \frac{3}{36} = \frac{1}{12}$$

and

$$P(X \geq 10) = \frac{6}{36} = \frac{1}{6}.$$

Inequalities and Cont. RVs

For continuous random variables whether or not the inequality is strict does not change the probability! If X is a continuous random variable, then

$$P(X > 5) = P(X \geq 5).$$

7 Normal Random Variables

7.1 Practice Computing Normal Probabilities

Use the *Normal Calculator* on StatCrunch to compute the normal probability. Sketch the appropriate area related to the probability.

1. A citrus farmer grows mandarin oranges. The diameter of the harvested oranges follow a normal distribution with a mean of 5.85 cm and a standard deviation of 0.24 cm. Find the probability that a random orange has a diameter that is at least 6.0 cms.
2. Using the same distribution from above, what is the probability that a random orange has a diameter between 3 and 4 cms?
3. The lifespan of a fitness tracking device follows a normal distribution with a mean of 4.1 years and a standard deviation of 1.8 years. What is the probability that a random fitness tracker lasts less than 2.5 years?
4. Suppose X follows a normal distribution with mean 14 and standard deviation 3.6. Find c such that $P(X > c) = 0.456$.

7.2 Z-Scores

For a random variable with any distribution, we can compute the **z-score**. To compute the z-score of a datapoint x we compute,

$$z = \frac{x - \mu}{\sigma}$$

where μ is the mean of the distribution and σ is the standard deviation. Z-scores are a useful tool for standardizing data and comparing variables from different distributions.

If the distribution is normal, then the z-score is a **standard normal** random variable. The standard normal distribution is a normal distribution with mean 0 and standard deviation 1.

Example 7.1. Scores on a standardized test are approximately normally distributed with mean 150 and standard deviation 20. What is the **z-score** of someone who scored a 164?

Answer

To compute the z-score we subtract the mean and divide by the standard deviation.

$$z = \frac{164 - 150}{20} = \frac{14}{20} = 0.7.$$

A z-score tells us how many standard deviations a random variable is away from the mean. In the above example, a test score of 164 is 0.7 standard deviations *above average*.

Z-score are useful to compare statistics that come from different distributions. Consider the following example,

Example 7.2 (Comparing Home Run totals across eras)**.** In 1920, Babe Ruth hit a record 54 home runs. In 2001, Barry Bonds hit a record 73 home runs. Who was a better hitter for their time (ignoring the steroid controversy of course)?

In 1920, the league mean was only 5 home runs with a standard deviation of 6, so Ruth's z-score was

$$z = \frac{54 - 5}{6} = 8.2.$$

In 2001, the league mean was 20 home runs with a standard deviation of 11, ao Bonds's z-score was

$$z = \frac{73 - 20}{11} = 4.8.$$

This is one argument that suggests that Ruth was the better hitter for his era. Using z-scores here is useful because it accounts for the fact that in 2001 there were a lot more home runs across the league, as opposed to in 1920.

Z-Scores for Any Distribution

We can compute a z-score even if our distribution is not normal in shape. However, we should be careful in relating this to probability or percentiles. In the above example, we know that Ruth was further from the mean (when measured in standard deviations), but we cannot put a specific value to how much better he was because we don't know what the distribution looks like.

Example 7.3. SAT and ACT scores are approximately normal. The SAT has a mean score of 1030 with a standard deviation of 210, while the ACT has a mean of 19.5 and a standard deviation of 6.

If Priya got a 1300 on the SAT and Marcus had an ACT score of 28, who performed better?

Answer

Priya has a z-score of 1.29 and Marcus has a z-score of 1.42, which means Marcus scored more standard deviations above the tests average.

7.3 Practice Problems

1. Scores on a stats exam have a mean of 72 and a standard deviation of 8. Find the z-score for a student who scored 84, and interpret it in context (with a sentence).

Answer

$$z = \frac{x - \mu}{\sigma} = \frac{84 - 72}{8} = 1.50$$

The student scored 1.5 standard deviations above the class mean.

2. Using the same exam, a student's z-score was -1.25. What was their score on the test?

Answer

$$-1.25 = \frac{x - 72}{8}$$

$$-1.25 \times 8 = x - 72$$

$$-10 = x - 72$$

$$x = 62$$

3. Ana scored 1350 on a test with mean 1050 and SD 200. Ben scored 28 on a different test with mean 21 and SD 5. Who did better relative to the test they took?

Answer

$$z_{\text{Ana}} = \frac{1350 - 1050}{200} = 1.50 \qquad z_{\text{Ben}} = \frac{28 - 21}{5} = 1.40$$

Ana did slightly better relative to her group, since her z-score is higher.

4. What does a z-score of 0 mean?

Answer

A z-score of 0 means the value equals the mean.

Let Z be a **standard normal random variable**.

5. Find $P(Z < 1.28)$.

Answer

Using StatCrunch's normal calculator: $P(Z < 1.28) = 0.8997$

6. Find $P(Z > -0.75)$.

Answer

$$P(Z > -0.75) = 0.7734$$

7. Find the value c such that $P(Z > c) = 0.05$.

Answer

$c = 1.645$.

8. Adult female heights are approximately normal with mean 64.5 in and SD 2.5 in. What proportion of women are taller than 68 in?

Answer

$$P(X > 68) = 0.0808$$

About 8.1% of women are taller than 68 in.

9. Using the same distribution, what proportion of women are between 60 and 67 inches tall?

Answer

$$P(60 < X < 67) = 0.8054$$

10. A battery's lifetime is normal with mean 500 hours and SD 40 hours. The manufacturer wants to set a warranty so that only 2% of batteries fail before it expires. How many hours should the warranty cover?

Answer

We want c such that, $P(X < c) = 0.02$. Using the normal calculator we get, $c = 417.8$ hours.

8 Sampling

8.1 Population Versus Sample

A **population** is the entire group of individuals/items we wish to investigate and draw conclusions about. It is often unreasonable or impossible to observe or measure all individuals of a population, so in practice we take a **sample** of individuals. In Topic 3, we addressed the importance of random sampling as a way to help draw fair conclusions.

Example 8.1. A city wants to determine the need for additional bus routes among city residents. Since it isn't practical to ask everyone, the transportation department randomly selects 800 households and asks how often they would use a new bus route if one were added near them.

- **Population:** all residents of the city.
- **Sample:** the 800 randomly selected households that were surveyed.

Example 8.2. A quality control engineer at a factory wants to know the average lifetime of the light bulbs produced on a particular assembly line. Testing a bulb means running it until it burns out, so testing every bulb would leave nothing to sell. Instead, the engineer pulls 50 bulbs at random from the week's production and records how many hours each one lasts.

- **Population:** all light bulbs produced on that assembly line.
- **Sample:** the 50 bulbs that were tested.

Example 8.3. A university wants to know how many hours per week its undergraduates spend working at paid jobs. The registrar's office randomly selects 300 students from the enrollment list and sends them a short survey.

- **Population:** all undergraduate students enrolled at the university.
- **Sample:** the 300 students selected to receive the survey.

The goal of taking a **sample** is to try to draw conclusions about the entire **population**, however we should not expect the sample to exactly reveal the underlying population behavior. Moreover, from sample to sample, we expect variation. This is known as **sampling variability**.

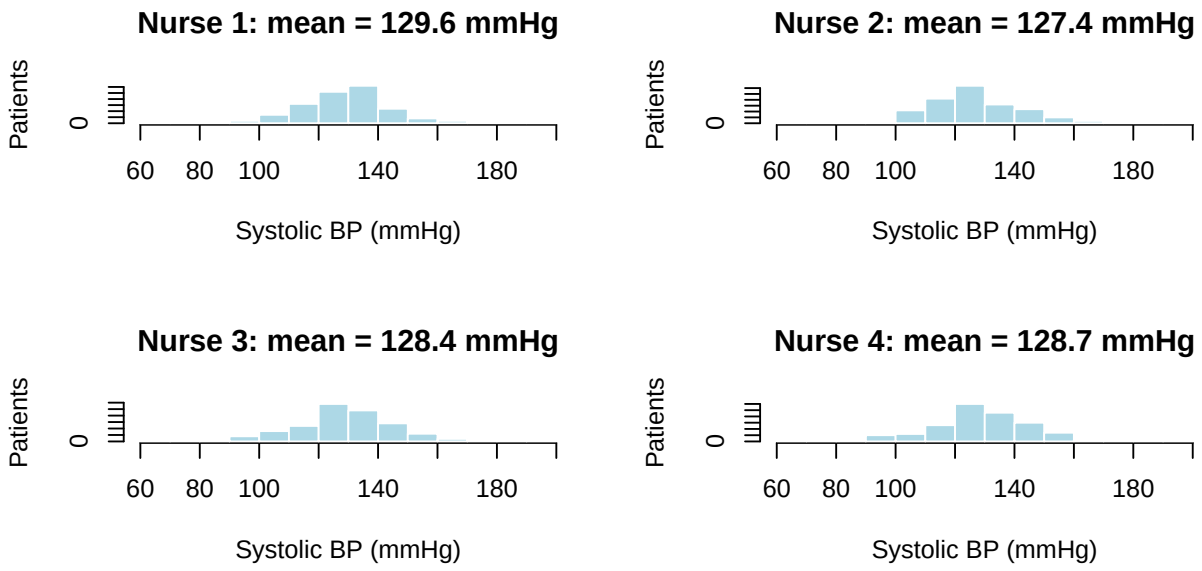


Figure 8.1: Records of patient BP by four different nurses.

Example 8.4. Four research nurses each record the systolic blood pressure of 100 randomly selected adults visiting the same hospital's outpatient clinic.

We see that each of the nurses records a slightly different mean and distribution. The population in this case is all the clinic's patients, however each nurse only sees a sample of that. This is **sample variability**.

A **parameter** is a number that describes the population, while a **statistic** is a number that describes a sample. We may not know parameters, but they have some underlying single value. A statistic, depends on the sample I take. In the above example, each nurse records a different mean. These means are *statistics* and they depend on the sample that was taken.

	Parameter (Population)	Statistic (Sample)
Proportion	p	$\hat{p}$
Mean	μ	$\bar{x}$
Standard Deviation	σ	s

A principle question we want to answer is *how close are the statistics to the parameters?* For example, how close are the nurse's means to the actual population mean (a parameter).

8.2 Sample Proportion

Example 8.5. A college wants to know what proportion of their most recent graduates have a job six months after graduating. The college randomly samples 500 graduates and finds that 64% of graduates in the sample have a job, so $\widehat{p} = 0.64$.

We can think of the sample proportion as a *random variable* that changes depending on the sample I take.

Example 8.6. Claudia and Joey want to know what proportion of their classmates have already finished Homework 4. Claudia samples 20 peers and finds that 32% of them have finished Homework 4. Joey samples a different 20 peers and finds that 39% of them have finished Homework 4.

If we collected a bunch of sample proportions from different random samples, the distribution of these values would be *approximately normal*. This is a result of the **Central Limit Theorem**.